ICMA216

Calculus IIIA

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Trimester 3, 2022-2023

Chapter 1

Calculus Review

Trigonometric Functions and Identities

Recall that

$$\sin(\pi - x) = \sin x, \quad \sin(\pi + x) = -\sin(x),$$

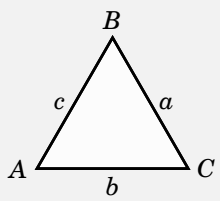
and that

$$\cos(\pi - x) = \cos(\pi + x) = -\cos(x).$$

We will also be making use of the following basic trigonometric identities

- $\cos^2\theta + \sin^2\theta = 1$
- $\sin 2x = 2\sin x \cos x$ [Double Angle Formula (sin)]
- $\cos 2x = \cos^2 x - \sin^2 x = 2\cos^2 x - 1 = 1 - 2\sin^2 x$ [Double Angle Formula [cos]]

Theorem 1 (Law of Sine and Cosine). *For the triangle ABC below*



The law of sine states that

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

The law of cosine states that

$$a^2 = b^2 + c^2 - 2bc\cos A, \quad b^2 = a^2 + c^2 - 2ac\cos B, \quad c^2 = a^2 + b^2 - 2ab\cos C$$

Differentiation

Trignometric Differentiation

$$\frac{d}{dx}\sin x = \cos x$$
$$\frac{d}{dx}\cos x = -\sin x$$
$$\frac{d}{dx}\tan x = \sec^2 x = \frac{1}{\cos^2 x}$$

Definition 1 (Product Rule).

$$\frac{d}{dx}[f(x)g(x)] = f(x)\frac{d}{dx}[g(x)] + g(x)\frac{d}{dx}[f(x)]$$

Example 1. Find the first derivative of $f(x) = (\sec x)(x + \tan x)$.

$$\begin{aligned} f'(x) &= (\sec x)(x+\tan x)' + (x+\tan x)(\sec x)' \\ &= \sec x(1+\sec^2 x) + (x+\tan x)(\sec x \tan x) \end{aligned}$$

Example 2. Find the first derivative of $f(x) = x^2 \cos x \sin x$

$$\begin{aligned} f'(x) &= x^2 \cos x \sin x \\ &= (x^2\cos x)(\sin x)' + (\sin x)(x^2 \cos x)' \\ &= (x^2 \cos x)(\cos x) + (\sin x)\left[x^2(\cos x)' + (\cos x)(x^2)'\right] \\ &= x^2\cos^2 x + (x^2 \sin x)(-\sin x) + (\sin x)(\cos x)(2x) \\ &= x^2\cos^2 x - x^2 \sin^2 x + 2x \sin x \cos x \end{aligned}$$

Definition 2 (Quotient Rule).

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)\frac{d}{dx}f(x) - f(x)\frac{d}{dx}g(x)}{(g(x))^2}$$

Example 3. Find $\left[\frac{f}{g}\right]'(5)$ given that $f(5) = -4, f'(5) = 2, g(5) = -7, g'(5) = -6$.

We know from the form of the quotient rule that it is

$$\begin{aligned}\left[\frac{f}{g}\right]'(5) &= \frac{g(5)\frac{d}{dx}f(5) - f(5)\frac{d}{dx}g(5)}{(g(5))^2} \\ &= \frac{(-7)(2)-(-4)(-6)}{(-7)^2} \\ &= \frac{-38}{49}\end{aligned}$$

Example 4. Find the derivative of $f(x) = \frac{(x-1)\sin x}{\sqrt{x+2}(x+4)^{3/2}}$

$$\begin{aligned}\ln f(x) &= \ln(x-1) + \ln[\sin x] - \frac{1}{2}\ln(x+2) - \frac{3}{2}\ln(x+4) \\ \frac{1}{f(x)}f'(x) &= \frac{1}{x-1} + \frac{\cos x}{\sin x} - \frac{1}{2}\cdot\frac{1}{x+2} - \frac{3}{2}\cdot\frac{1}{x+4} \\ f'(x) &= \left[\frac{(x-1)\sin x}{\sqrt{x+2}(x+4)^{3/2}}\right]\left[\frac{1}{x-1} + \frac{\cos x}{\sin x} - \frac{1}{2}\cdot\frac{1}{x+2} - \frac{3}{2}\cdot\frac{1}{x+4}\right]\end{aligned}$$

Example 5. Suppose $y = e^{2021x}$, what if we wanted to find the nth derivative of this.

$$y' = (2021)^n e^{2021x}$$

Definition 3 (Chain Rule). Let $y = f(g(x))$ and $u = g(x)$, then $y = f(u)$ and

$$\frac{dy}{dx} = \frac{dy}{du}\cdot\frac{du}{dx}$$

Integration

Definition 4 (u-substitution).

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du = F(u) + C$$

where $u = g(x)$ and $du = g'\,dx$.

Example 6. $\int[\ln(x^2) - 2\ln(2x)]\,dx$

$$\begin{aligned}\int \left[\ln(x^2) - 2\ln(2x)\right]\,dx &= \int \ln\left(\frac{x^2}{(2x)^2}\right)\,dx \\ &= \int \ln\left(\frac{1}{4}\right)\,dx \\ &= x\ln\left(\frac{1}{4}\right) + C \\ &= x(\ln 1 - \ln 4) + C \\ &= -x\ln 4\end{aligned}$$

Example 7. $\int \frac{\cos 2x}{\cos x + \sin x}\,dx$

$$\begin{aligned}\int \frac{\cos 2x}{\cos x + \sin x}\,dx &= \int \frac{\cos^2 x - \sin^2 x}{\cos x + \sin x}\,dx \\ &= \int \frac{(\cos x - \sin x)(\cos x + \sin x)}{\cos x - \sin x}\,dx \\ &= \int \cos x - \sin x\,dx \\ &= \sin x + \cos x + C\end{aligned}$$

Example 8. $\int \frac{x^2}{x^2+1}\,dx$

$$\begin{aligned}\int \frac{x^2}{x^2+1}\,dx &= \int \frac{x^2+1-1}{x^2+1}\,dx \\ &= \int \left(1 - \frac{1}{x^2+1}\right)\,dx \\ &= x - \tan^{-1} x + C\end{aligned}$$

Example 9. $\int \frac{\cos\left(\frac{\sin x}{\cos x}\right)}{\cos x \cos x}\,dx$

For this, we let $u = \tan x, du = \sec^2 x\,dx = \frac{1}{\cos^2 x}\,dx$. So

$$\begin{aligned}\int \frac{\cos\left(\frac{\sin x}{\cos x}\right)}{\cos x \cos x}\,dx &= \int \cos u\,du \\ &= \sin u + C \\ &= \sin(\tan x) + C\end{aligned}$$

Example 10. $\int(\tan x)\ln(\cos x)\,dx$

Let $u = \ln(\cos x), du = -\frac{\sin x}{\cos x}\,dx = -\tan x\,dx$.

$$\begin{aligned}\int (\tan x)\ln(\cos x)\,dx &= -\int u\,du \\ &= -\frac{u^2}{2} + C \\ &= -\frac{\ln^2(\cos x)}{2} + C\end{aligned}$$

Example 11. $\int \frac{3+x}{1+x^2}\,dx$

$$\begin{aligned}\int \frac{3+x}{1+x^2}\,dx &= \int \frac{d}{1+x^2}\,dx + \int \frac{x}{1+x^2}\,dx \\ &= 3\tan^{-1}x + \frac{1}{2}\int \frac{1}{u}\,du, \quad u = 1+x^2 \\ &= 3\tan^{-1}x + \frac{1}{2}\ln|u| + C \\ &= 3\tan^{-1}x + \frac{1}{2}\ln\left(1+x^2\right) + C\end{aligned}$$

Definition 5 (Integration By Parts).

$$\int u\,dv = uv - \int v\,du$$

The method of choosing functions integration by parts is usually $LIATE$: logarithmic functions, inverse trigonometric functions, algebraic, trigonometric, exponential.

Example 12. $\int \frac{x^2+1}{x(x+1)^2}\,dx$

First perform partial fraction decomposition

$$\begin{aligned}\frac{x^2+1}{x(x+1)^2} &= \frac{A}{x} + \frac{B}{x+1} + \frac{C}{(x+1)^2} \\ x^2+1 &= A(x+1)^2 + Bx(x+1) + Cx\end{aligned}$$

From this, we determine that when $x = 0$, $A = 1$ and when $x = -1$, $C = -2$. For B, we can plug in the values of A and C obtained to get

$$\begin{aligned}&x^2+1 = (x^2+2x+1) + B(x^2+x) - 2x \\ &B+1 = 0 \implies B = 0\end{aligned}$$

Returning to our integral, we have

$$\begin{aligned}\int \frac{x^2+1}{x(x+1)^2}\,dx &= \int \frac{1}{x} - \frac{2}{(x+1)^2}\,dx \\ &= \ln|x| + \frac{2}{x+1} + C\end{aligned}$$

Chapter 2

Rectangular Coordinates in 3-Space; Vectors

Consider movement in one dimension. This would just consist of points on a number line, on which we would be able to move left and right.

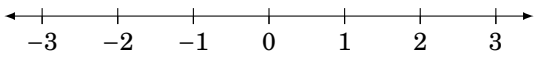


Figure 2.1: One dimensional space.

What about a two-dimensional space? We have a bit more freedom in traveling now – we can move both horizontally and vertically.

Three Dimensional Space

Consider the following set of points

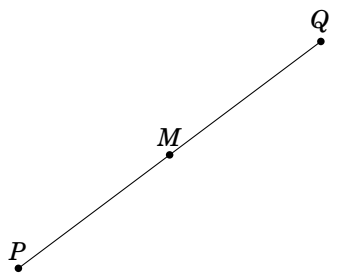


Figure 2.2: Midpoint between two points

Definition 6 (Midpoint). The midpoint between $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ is

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2}\right)$$

Definition 7 (Distance from Origin). The distance from the origin to the point $P(x_0, y_0, z_0)$ is

$$d = \sqrt{x_0^2 + y_0^2 + z_0^2}$$

Definition 8 (Distance Between Vectors). The distance from P_0 to another point P_1 is

$$d = \sqrt{(x_1 - x_0)^2 + (y_1 - y_0)^2 + (z_1 - z_0)^2}$$

Example 13. Find the distance d between the points $(1, -2, 4)$ and $(-3, -6, -3)$.

$$\begin{aligned} d &= \sqrt{(1-(-3))^2 + (-2-(-6))^2 + (4-(-3))^2} \\ &= \sqrt{(4)^2 + (4)^2 + (7)^2} \\ &= 9 \end{aligned}$$

Surfaces

Planes

A plane has the general form

$$Ax + By + Cz = D$$

For example, $x = 0, x = 3$ or $z = 5$. In a two-dimensional setting, this would be a line. However, in a three-dimensional setting, this would be an entire cross-section* of the axes (a plane).

Cylindrical Surfaces

A cylindrical surface might be something like

$$x^2 = y + 3$$

In a two-dimensional space, this would look like a parabola. Converting this over to a three-dimensional representation, we obtain a cylindrical surface.

Spheres

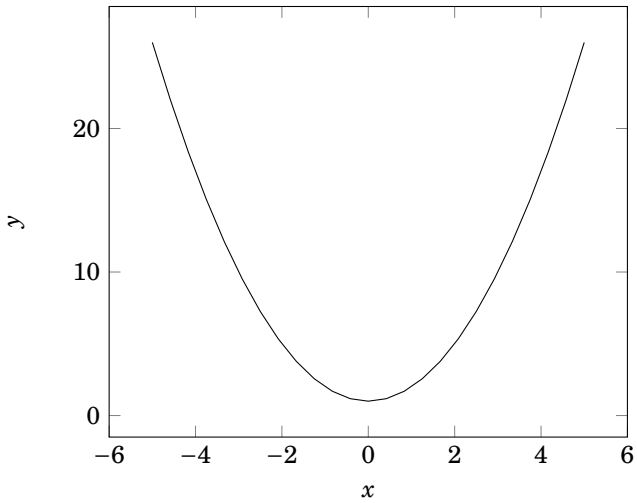
A sphere has the general form

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = r^2$$

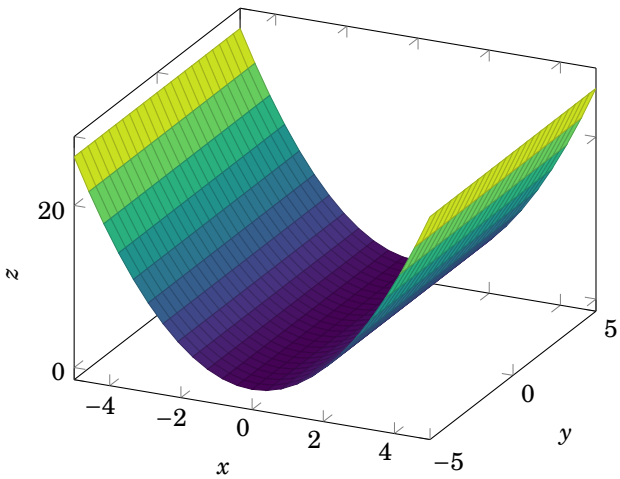
where (x_0, y_0, z_0) are the center of the sphere and its radius is r.

Example 14. [Sketching] *Sketch* $y = x^2 + 1$ *in* $\mathbb{R}^3$.

In two dimensions, we have



In three dimensions, we have the following



Sketch $x^2 + z^2 - 8z + 12 = 0$ *in* $\mathbb{R}^3$

First we will have to complete the square

$$\begin{aligned} x^2 + z^2 - 8z + 12 &= 0 \\ x^2 + (z-4)^2 &= -2 + 16 \\ x^2 + (z-4)^2 &= 4 \end{aligned}$$

In two dimensions, we have a circle. In three-dimensions, we have a tube that extends infinitely along the z-dimension. The intersection* of this tube on the xz-plane is a circle centered at $(0,0,4)$ of radius 2 with equation $x^2 + (z-4)^2 = 4$.

What about the intersection with the yz-plane? We consider when $x = 0$. Then

$$\begin{aligned} (0)^2 + (z-4)^2 &= 4 \\ z - 4 &= \pm 2 \\ z &= 6, 2 \end{aligned}$$

Thus we obtain two lines $z = 6$ and $z = 2$.

What about the intersection* with the xy plane? In this instance, $z = 0$, so we have

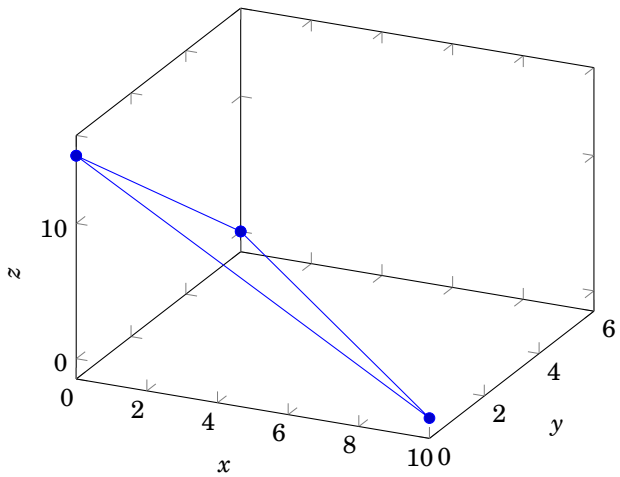
$$\begin{aligned} x^2 + (0-4)^2 &= 4 \\ x^2 + 16 &= 4 \\ x^2 &= -12 \end{aligned}$$

This cannot occur, which means that this is nowhere on the xy plane.

Example 15. For the equation

$$3x + 5y + 2z = 30$$

Find the points where this surface intersects the cooordinates axes. We can begin with a sketch.



Consider when x, y and z are zero, respectively.

x	y	z
0	0	15
0	6	0
10	0	0

Example 16. Write the equation of a sphere centered at $(3, 4, -1)$ with radius 6.

$$\begin{aligned} (x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 &= r^2 \\ (x-3)^2 + (y-4)^2 + (z+1)^2 &= 36 \end{aligned}$$

Vectors

Vectors have both length and direction (with no actual initial point). We use the notation $\hat{\imath}, \hat{\jmath}, \hat{k}$ as a basis for the vector in 3-dimensional space. $\hat{\imath}$ is the unit vector in the positive x-direction, $\hat{\jmath}$ is the

unit vector in the positive y-direction and $\hat{k}$ is the unit vector in the positive z direction.

$$\hat{i} = \langle 1,0,0\rangle$$
$$\hat{j} = \langle 0,1,0\rangle$$
$$\hat{k} = \langle 0,0,1\rangle$$

We can represent vectors as follows

$$\langle 3,5,2\rangle = 3\hat{i} + 5\hat{j} + 2\hat{k}$$

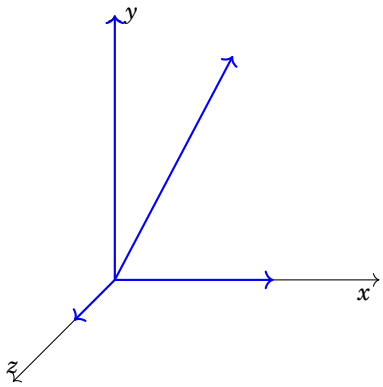


Figure 2.3: $\vec{v} = \langle 3,5,2\rangle$

One important vector is the zero-vector

$$\vec{0} = \langle 0,0,0\rangle$$

To find a vector between two vectors $P(x_1,y_1,z_1)$ and $Q(x_2,y_2,z_2)$, we find

$$\langle x_2 - x_1, y_2 - y_1, z_2 - z_1\rangle$$

Theorem 2 (Arithmetic Operations on Vectors). *If $\vec{v} = \langle v_1,v_2,v_3\rangle$ and $\vec{w} = \langle w_1,w_2,w_3\rangle$, then*

1. $\vec{v} + \vec{w} = \langle v_1 + w_1, v_2 + w_2, v_3 + w_3\rangle$
2. $\vec{v} - \vec{w} = \langle v_1 - w_1, v_2 - w_2, v_3 - w_3\rangle$
3. $k\vec{v} = \langle kv_1, kv_2, kv_3\rangle$

Theorem 3 (Rules of Vector Arithmetic). *Some properties of vector arithmetic include the following*

1. $\vec{v} + \vec{w} = \vec{w} + \vec{v}$

Proof.

$$\begin{aligned}\vec{v}+\vec{w} &= \langle v_1, v_2\rangle + \langle w_1, w_2\rangle \\ &= \langle v_1+w_1, v_2+w_2\rangle \\ &= \langle w_1+v_1, w_2+v_2\rangle \\ &= \langle w_1, w_2\rangle + \langle v_1, v_2\rangle \\ &= \vec{w}+\vec{v}\end{aligned}$$

□

2. $\vec{v}+(-\vec{v})=\vec{0}$
3. $\vec{w}=c\vec{v}, c\neq 0 \iff \vec{w} \parallel \vec{v}$[a]

[a] $\vec{w}$ is parallel to $\vec{v}$

Definition 9 (Norm). A vector $\vec{v}=\langle v_1, v_2, v_3\rangle$ has the norm (or length)

$$\|\vec{v}\| = \sqrt{v_1^2+v_2^2+v_3^2}$$

The norm of a vector has the following properties

1. $\|c\vec{v}\| = |c|\|\vec{v}\|$
2. The unit vector (length one) of $\vec{v}$ is $\frac{\vec{v}}{\|\vec{v}\|}$

Example 17. *Find a vector of length 2 that makes an angle of $\frac{\pi}{4}$ with the positive x-axis.*

Let $\vec{v}$ be such a vector. We can write it in terms of the unit vector multiplied by some constant.

$$\vec{v} = c\langle 1,1\rangle$$

We can calculate the norm

$$\begin{aligned}\|\vec{v}\| &= \|c\langle 1,1\rangle\| \\ &= c\sqrt{2}\end{aligned}$$

Since we want the norm to be two

$$c = \frac{\|\vec{v}\|}{\sqrt{2}} = \frac{2}{\sqrt{2}}$$

Example 18. *Find the angle that the vector $\vec{v}=-\sqrt{3}\hat{i}+\hat{j}$ makes with the positive axis.*

It makes the angle $\frac{\pi}{6}$ with the x-axis in 2nd quadrant, so that means the angle it makes with the positive axis is

$$\theta = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$$

Example 19. Let $\vec{v}=\langle 1,2,4\rangle$ and let $\vec{w}=\langle 2,-4,1\rangle$.

1. Find $3\vec{v} - 2\vec{w}$.

$$\begin{aligned} 3\vec{v} - 2\vec{w} &= 3\langle 1,2,4\rangle - 2\langle 2,-4,1\rangle \\ &= \langle 3,6,12\rangle - \langle 4,-8,2\rangle \\ &= \langle -1,14,10\rangle \end{aligned}$$

2. Find a vector of length 3 in the opposite direction of $\vec{v}$.

   To find a vector in the opposite direction of $\vec{v}$, we can simply negate it. However, to obtain the length, we will have to solve for some constant. Let's check the norm of $\vec{v}$.

$$\begin{aligned} \|\vec{v}\| &= \sqrt{1^2 + 2^2 + 4^2} \\ &= \sqrt{21} \end{aligned}$$

   We can normalize $\vec{v}$ to a unit vector and multiply it by -3 to obtain our desired vector

$$-3 \cdot \frac{\vec{v}}{\sqrt{21}} = -\frac{3}{\sqrt{21}}\langle 1,2,4\rangle$$

Dot Product and Projections

Definition 10 (Dot Product). *Let* $\mathbf{u} = \langle u_1, u_2\rangle$ *and* $\mathbf{v} = \langle v_1, v_2\rangle$ *be vectors in two-dimensional space. Then their dot product, denoted* $\mathbf{u}\cdot\mathbf{v}$ *is defined as*

$$\mathbf{u}\cdot\mathbf{v} = u_1v_1 + u_2v_2.$$

Similarly, if $\mathbf{u} = \langle u_1, u_2, u_3\rangle$ *and* $\mathbf{v} = \langle v_1, v_2, v_3\rangle$*, then*

$$\mathbf{u}\cdot\mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3.$$

Theorem 4 (Properties of Dot Product). The dot product has the following properties

1. $\mathbf{u}\cdot\mathbf{v} = \mathbf{v}\cdot\mathbf{u}$

   Proof.

$$\begin{aligned} \mathbf{u}\cdot\mathbf{v} &= \langle u_1, v_1\rangle \cdot \langle u_2, v_2\rangle \\ &= u_1v_1 + u_2v_2 \\ &= u_2v_2 + u_1v_2 \\ &= \mathbf{v}\cdot\mathbf{u} \end{aligned}$$

   □

2. $\mathbf{v}\cdot\mathbf{v} = \|\mathbf{v}\|^2$

Proof.

$$\begin{aligned}\mathbf{v}\cdot\mathbf{v} &= \langle v_1, v_2\rangle \cdot \langle v_1, v_2\rangle \\ &= v_1 v_2 + v_2 v_2 \\ &= v_1^2 + v_2^2 \\ &= \sqrt{v_1^2 + v_2^2}^2 \\ &= \|\mathbf{v}\|^2\end{aligned}$$

□

3. $\mathbf{0}\cdot\mathbf{v} = 0$

   Proof. Trivial. □

4. $\mathbf{u}\cdot(\mathbf{v}+\mathbf{w}) = \mathbf{u}\cdot\mathbf{v} + \mathbf{u}\cdot\mathbf{w}$

Theorem 5. *Let θ be the angle between $\mathbf{u}$ and $\mathbf{v}$. Then*

$$\cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}.$$

Proof. Consider $\|\mathbf{v}-\mathbf{u}\|^2$, first by applying the law of cosines, $c^2 = a^2 + b^2 - 2ab\cos\theta$:

$$\begin{aligned}\|\mathbf{v}-\mathbf{u}\|^2 &= \|\mathbf{v}+(-\mathbf{u})\|^2 \\ &= \|\mathbf{v}\|^2 + \|-\mathbf{u}\|^2 - 2\|\mathbf{v}\|\|-\mathbf{u}\|\cos\theta \\ &= \|\mathbf{v}\|^2 + \|-\mathbf{u}\|^2 - 2\|\mathbf{v}\|\|\mathbf{u}\|\cos\theta.\end{aligned}$$

And next by considering property 4 of the dot product

$$\begin{aligned}\|\mathbf{v}-\mathbf{u}\|^2 &= (\mathbf{v}-\mathbf{u})\cdot(\mathbf{v}-\mathbf{u}) \\ &= \mathbf{v}\cdot\mathbf{v} - \mathbf{v}\cdot\mathbf{u} - \mathbf{u}\cdot\mathbf{v} + \mathbf{u}\cdot\mathbf{u} \\ &= \mathbf{v}\cdot\mathbf{v} - 2\mathbf{v}\cdot\mathbf{u} + \mathbf{u}\cdot\mathbf{u} \\ &= \|\mathbf{v}\|^2 - 2\mathbf{v}\cdot\mathbf{u} + \|\mathbf{u}\|^2.\end{aligned}$$

Therefore, it must be the case that

$$\begin{aligned}2\mathbf{v}\cdot\mathbf{u} &= 2\|\mathbf{v}\|\|\mathbf{u}\|\cos\theta \\ \mathbf{v}\cdot\mathbf{u} &= \|\mathbf{v}\|\|\mathbf{u}\|\cos\theta \\ \cos\theta &= \frac{\mathbf{v}\cdot\mathbf{u}}{\|\mathbf{v}\|\|\mathbf{u}\|}.\end{aligned}$$

□

Example 20. Find the angle between the vector $\mathbf{u} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and

(a) $\mathbf{v} = 3\mathbf{i} - \mathbf{k}$

$$\begin{aligned}
\cos\theta &= \frac{\mathbf{v}\cdot\mathbf{u}}{\|\mathbf{v}\|\,\|\mathbf{u}\|} \\
&= \frac{\langle 1,2,-1\rangle \cdot \langle 3,0,-1\rangle}{\sqrt{1^2+2^2+(-1)^2}\cdot\sqrt{3^2+0^2+(-1)^2}} \\
&= \frac{4}{\sqrt{6}\cdot\sqrt{10}} \\
&= \frac{4}{\sqrt{60}} \\
&= \frac{2}{\sqrt{15}} \\
\theta &= \cos^{-1}\left(\frac{2}{\sqrt{15}}\right)
\end{aligned}$$

(b) $\mathbf{w} = -2\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}$

$$\begin{aligned}
\cos\theta &= \frac{\mathbf{v}\cdot\mathbf{u}}{\|\mathbf{v}\|\,\|\mathbf{u}\|} \\
&= \frac{\langle 1,2,-1\rangle \cdot \langle -2,-4,2\rangle}{\sqrt{1^2+2^2+(-1)^2}\cdot\sqrt{(-2)^2+(-4)^2+2^2}} \\
&= \frac{-12}{\sqrt{6}\cdot\sqrt{24}} \\
&= -\frac{12}{\sqrt{144}} \\
&= -1 \\
\theta &= \cos^{-1}(-1) \\
&= \pi
\end{aligned}$$

Definition 11 (Orthogonal Vectors). *The (non-zero) vectors* $\mathbf{u}$ *and* $\mathbf{v}$ *are orthogonal or perpendicular* $\left(\theta = \frac{\pi}{2}\right)$ *if and only if* $\mathbf{u}\cdot\mathbf{v} = 0$.

Cross Product

Definition 12 (Determinant). The determinant for a 2-by-2 matrix is the following

$$\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} = a_1 b_2 - a_2 b_1.$$

For a 3-by-3 matrix

$$\begin{aligned}
\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} &= a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix} \\
&= a_1(b_2c_3 - b_3c_2) - a_2(b_1c_3 - b_3c_1) + a_3(b_1c_2 - b_2c_1).
\end{aligned}$$

Definition 13 (Cross Product). *Let* $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ *and* $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$. *Then, the cross product between the two is*

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}.$$

Example 21. Find $\mathbf{i} \times \mathbf{j}$

$$\begin{aligned} \mathbf{i} \times \mathbf{j} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{j} \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} \\ &= \mathbf{i} \begin{vmatrix} 0 & 0 \\ 1 & 0 \end{vmatrix} - \mathbf{j} \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} + \mathbf{k} \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \\ &= \mathbf{i} \cdot 0 - \mathbf{j} \cdot 0 + \mathbf{k} \cdot 1 \\ &= \mathbf{k} \\ &= \langle 0, 0, 1 \rangle. \end{aligned}$$

Example 22. Let $\mathbf{u} = \langle 1, -2, 2 \rangle$, $\mathbf{v} = \langle 3, -1, 1 \rangle$ and $\mathbf{w} = \langle 1, 0, -2 \rangle$. Find

1. $\mathbf{u} \times \mathbf{w}$

$$\begin{aligned} \mathbf{u} \times \mathbf{w} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -2 & 2 \\ 1 & 0 & -2 \end{vmatrix} \\ &= \mathbf{i} \begin{vmatrix} -2 & 2 \\ 0 & -2 \end{vmatrix} - \mathbf{j} \begin{vmatrix} 1 & 2 \\ 1 & -2 \end{vmatrix} + \mathbf{k} \begin{vmatrix} 1 & -2 \\ 1 & 0 \end{vmatrix} \\ &= 4\mathbf{i} + 4\mathbf{j} + 2\mathbf{k} \end{aligned}$$

2. $\mathbf{w} \times \mathbf{u}$

$$\begin{aligned} \mathbf{w} \times \mathbf{u} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & -2 \\ 1 & -2 & 2 \end{vmatrix} \\ &= \mathbf{i} \begin{vmatrix} 0 & -2 \\ -2 & 2 \end{vmatrix} - \mathbf{j} \begin{vmatrix} 1 & -2 \\ 1 & 2 \end{vmatrix} + \mathbf{k} \begin{vmatrix} 1 & 0 \\ 1 & -2 \end{vmatrix} \\ &= -4\mathbf{i} - 4\mathbf{j} - 2\mathbf{k} \end{aligned}$$

3. $\mathbf{v} \times \mathbf{v}$

$$\begin{aligned} \mathbf{v} \times \mathbf{v} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -1 & 1 \\ 3 & -1 & 1 \end{vmatrix} \\ &= \mathbf{i} \begin{vmatrix} -1 & 1 \\ -1 & 1 \end{vmatrix} - \mathbf{j} \begin{vmatrix} 3 & 1 \\ 3 & 1 \end{vmatrix} + \mathbf{k} \begin{vmatrix} 3 & -1 \\ 3 & -1 \end{vmatrix} \\ &= \mathbf{i} \cdot 0 - \mathbf{j} \cdot 0 + \mathbf{k} \cdot 0 \\ &= 0 \end{aligned}$$

Theorem 6 (Properties of Cross Product).

1. $\mathbf{u} \times \mathbf{v} = -\mathbf{v} \times \mathbf{u}$.
2. $\mathbf{u} \times \mathbf{v} = 0 \iff \mathbf{u} \parallel \mathbf{v}$.
3. $\mathbf{u} \times \mathbf{v}$ is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$. In other terms.
$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{u} = 0, \quad (\mathbf{u} \times \mathbf{v}) \cdot \mathbf{v} = 0.$$
4. $\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| \sin\theta$, which is the area of a parallelogram.

Projections

Definition 14 (Scalar Projection). The *scalar projection* of a vector $\mathbf{b}$ onto $\mathbf{a}$ is defined to be
$$\text{comp}_{\mathbf{a}}\mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|} = \frac{\|\mathbf{a}\| \|\mathbf{v}\| \cos\theta}{\|\mathbf{a}\|}$$

Definition 15. The *vector projection* of a vector $\mathbf{b}$ onto $\mathbf{a}$ is defined as
$$\text{proj}_{\mathbf{a}}\mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|} \cdot \frac{\mathbf{a}}{\|\mathbf{a}\|}$$

Example 23. *Find vector and scalar projections of* $\mathbf{m} = \langle 2, 1, 5 \rangle$ *onto* $\mathbf{n} = \langle 1, 2, 3 \rangle$.

The scalar projection of $\mathbf{m}$ onto $\mathbf{n}$ is the following
$$\begin{aligned} \text{comp}_{\mathbf{n}}\mathbf{n} &= \frac{\mathbf{m} \cdot \mathbf{n}}{\|\mathbf{n}\|} \\ &= \frac{\langle 2, 1, 5 \rangle \cdot \langle 1, 2, 3 \rangle}{\sqrt{1^2 + 2^2 + 3^2}} \\ &= \frac{19}{\sqrt{14}}. \end{aligned}$$

As for the vector projection
$$\begin{aligned} \text{proj}_{\mathbf{n}}\mathbf{m} &= \frac{\mathbf{m} \cdot \mathbf{n}}{\|\mathbf{n}\|^2} \cdot \mathbf{n} \\ &= \frac{19}{14} \langle 1, 2, 3 \rangle \\ &= \left\langle \frac{19}{14}, \frac{38}{14}, \frac{57}{14} \right\rangle. \end{aligned}$$

Definition 16 (Scalar Triple Product). Let $\mathbf{a}, \mathbf{b}, \mathbf{c}$ be vectors. The *scalar triple product*

between the three is defined to be

$$\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

This also gives us the area of a *parallelepiped* – a three-dimensional shape whose faces are all parallelograms.

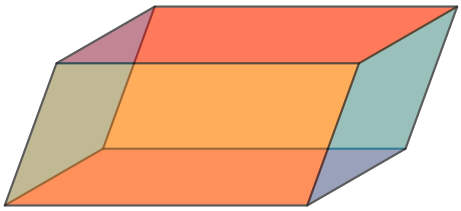


Figure 2.4: A parallelepiped [source].

Example 24. [Coplanar Vectors] *Suppose we are given vectors* $\mathbf{a} = \langle 1,2,3\rangle, \mathbf{b} = \langle 4,5,6\rangle$*, and* $\mathbf{c} = \langle 2,7,5\rangle$*. Are these vectors coplanar (i.e. do they lie in the same plane)?*

We can compute the scalar triple product and see if it is equal to zero.

$$\begin{aligned}
\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}) &= \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 2 & 7 & 5 \end{vmatrix} \\
&= 1\begin{vmatrix} 5 & 6 \\ 7 & 5 \end{vmatrix} - 2\begin{vmatrix} 4 & 6 \\ 2 & 5 \end{vmatrix} + 3\begin{vmatrix} 4 & 5 \\ 2 & 7 \end{vmatrix} \\
&= 1(25-42) - 2(20-12) + 3(28-10) \\
&= 1\cdot(-17) - 2\cdot 8 + 3\cdot 18 \\
&= 21.
\end{aligned}$$

As this is not equal to zero, $\mathbf{a}, \mathbf{b}$ and $\mathbf{c}$ are not coplanar.

Parametric Equations of Lines

Traditionally in a two-dimensional setting, the equation for a line is

$$y = ax + b.$$

However, in the three dimensional space, this would give us a plane. What if we just wanted a line? In a three dimensional space, lines can be written in *vector equation form*

$$\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}, \quad \langle x,y,z\rangle = \langle x_0,y_0,z_0\rangle + t\langle a,b,c\rangle,$$

where $\mathbf{r}_0$ is a fixed vector, t is some parameter and $\mathbf{v}$ is the direction vector. They can also be written in *parametric equation form*

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct.$$

Example 25. *Find a vector equation and parametric equation for the line that passes through the point* $(5,1,3)$ *and is parallel to the vector* $\mathbf{i}+4\mathbf{j}-2\mathbf{k}$.

In this instance, we see that we have a fixed point

$$\mathbf{r}_0 = \langle x_0, y_0, z_0\rangle = \langle 5,1,3\rangle,$$

and direction vector $\langle 1,4,-2\rangle$, so we have the vector equation

$$\langle x,y,z\rangle = \langle 5,1,3\rangle + t\langle 1,4,-2\rangle.$$

As for the parametric equation

$$x = 5+t, \quad y = 1+4t, \quad z = 3-2t.$$

Example 26. *Find a parametric equation for the line that passes through the point* $(0,-2,1)$ *and* $(1,1,1)$.

In this instance, we first need to form a direction vector from these points. Let $P = (0,-2,1)$ and $Q = (1,1,1)$. Then

$$\begin{aligned}\mathbf{r}_0 &= \overrightarrow{PQ}\\ &= \langle 1,1,1\rangle - \langle 0,-2,1\rangle\\ &= \langle 1,3,0\rangle.\end{aligned}$$

Thus we have the vector equation

$$\langle x,y,z\rangle = \langle 0,-2,1\rangle + t\langle 1,3,0\rangle,$$

and parametric equation

$$x = t, \quad y = -2+3t, \quad z = 1$$

Definition 17 (Parallel Lines). Given two vectors $\mathbf{r}_a = \mathbf{r}_0 + t\mathbf{v}_0$ and $\mathbf{r}_b = \mathbf{r}_1 + s\mathbf{v}_1$, then

$$\mathbf{r}_a \parallel \mathbf{r}_b \iff \mathbf{v}_0 \parallel \mathbf{v}_1 \iff \mathbf{v}_0 = c\mathbf{v}_1,$$

for some constant c.

Definition 18 (Intersecting Lines). In order to determine the intersection* point of two lines, we need merely equate them and solve for a common point. For example, suppose we wanted to find the intersecting point between two lines $\mathbf{r}_1 = \langle 1,1,0\rangle + t\langle 1,-1,2\rangle$ and $\mathbf{r}_2 = \langle 0,0,2\rangle + s\langle -1,1,0\rangle$. Let us first determine the parametric form for both.

$$\begin{aligned}\mathbf{r}_1 &: x = 1+t, \quad y = 1-t, \quad z = 2t\\ \mathbf{r}_2 &: x = 2-s, \quad y = s, \quad z = 2\end{aligned}$$

We now have a system of three equations and two variables

$$\begin{aligned} 1+t &= 2-s \\ 1-t &= s \\ 2t &= 2. \end{aligned}$$

From this, we see that $t=1$ and $s=0$. Plugging $t=1$ or $s=0$ into $\mathbf{r}_1$ or $\mathbf{r}_2$, respectively, gives us the point of intersection*

$$(x,y,z)=(2,0,2).$$

Definition 19 (Skew Lines). Skew lines are two lines that do not intersect and are not parallel. Is this possible? In two-dimensional space, it is not. However, in three dimensions, it most certainly is.

Definition 20 (Symmetric Equations). Note that there exists another form for lines with no extra variable t. Given

$$\langle x,y,z\rangle = \langle x_0,y_0,z_0\rangle + t\langle a,b,c\rangle,$$

we can rewrite in parametric form

$$x=x_0+at, \quad y=y_0+bt, \quad z=z_0+ct.$$

Rearranging for t in each term, we obtain the *symmetric equation*

$$t=\frac{x-x_0}{a}=\frac{y-y_0}{b}=\frac{z-z_0}{c}.$$

Furthermore, note that there is no unique way to write a straight line since the parameter $t\in(-\infty,\infty)$ can be shifted by any constant.

Example 27. [Skew Lines] Determine whether the following two lines are skew lines

$$\begin{aligned} L_1 &: \frac{x+2}{3}=\frac{y-5}{-4}=\frac{1-z}{2} \\ L_2 &: x=1-t, y=3+2t, z=-12-3t \end{aligned}$$

In order to determine whether or not two lines are skew lines, we need to find out if they are not parallel and if they do not intersect. First let us determine the vectors. L_2 has the direction vector $\mathbf{v}_2=\langle -1,2,-3\rangle$. For L_2, we have to rearrange to determine the parameter (say s).

$$\begin{aligned} \frac{x+2}{3}=s &\iff x=-2+3s \\ \frac{y-5}{-4}=s &\iff y=5-4s \\ \frac{1-z}{2}=s &\iff z=1-2s. \end{aligned}$$

Thus, L_1 has the direction vector $\mathbf{v}_1=\langle 3,-4,2\rangle$. In order to determine if they are parallel, i.e.,

$$\langle 3,-4,2\rangle = k\langle -1,2,-3\rangle,$$

we have to solve the system of equations

$$\begin{aligned} 3 &= -k \\ -4 &= 2k \\ 2 &= -3k. \end{aligned}$$

There is no solution to this system of equations, so $\mathbf{v}_1 \nparallel \mathbf{v}_2$, and hence $L_1 \nparallel L_2$. Next we must determine if the lines are intersecting. We can do so by equating the two lines

$$\begin{aligned} 1 - t &= -2 + 3s \\ 3 + 2t &= 5 - 4s \\ -12 - 3t &= 1 - 2s \end{aligned}$$

Solving this system of equations, we obtain $s = 2$ and $t = -3$. So from L_2

$$\begin{aligned} x &= (1 - (-3)) = 4 \\ y &= 3 + 2(-3) = -3 \\ z &= -12 - 3(-3) = -3 \end{aligned}$$

So the intersection* point is

$$(x, y, z) = (4, -3, -3),$$

and we conclude that L_1 and L_2 are not skew lines.

Planes

A vector $\mathbf{n}$ that is perpendicular to a plane is called a *normal vector*. Therefore, any $\mathbf{n}$ perpendicular to any vector $\mathbf{r} - \mathbf{r}_0$, where $\mathbf{r}$ and $\mathbf{r}_0$ are any two points on the plane. The equation of a plane is

$$\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0,$$

which can also be written in the form

$$ax + by + cz = d, \quad a, b, c, d \in \mathbb{R},$$

given that $\mathbf{n} = \langle a, b, c \rangle$. Why?

$$\begin{aligned} \mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) &= \langle a, b, c \rangle \cdot (\langle x, y, z \rangle - \langle x_0, y_0, z_0 \rangle) \\ &= \langle a, b, c \rangle \cdot \langle x - x_0, y - y_0, z - z_0 \rangle \\ &= a(x - x_0) + b(y - y_0) + c(z - z_0) \\ &= ax - ax_0 + by - by_0 + cz - cz_0 \\ ax + by + cz &= \underbrace{ax_0 + by_0 + cz_0}_{\text{constant}} \end{aligned}$$

As long as we are given $\mathbf{n}$ and a point $\mathbf{r}_0$, we can determine the equation of a plane.

Example 28. Let $\mathbf{n} = \langle 2,-2,1\rangle$ and $\mathbf{r}_0 = \langle -3,-4,5\rangle$. Find the equation of the plane.

$$\begin{aligned}\mathbf{n}\cdot(\mathbf{r}-\mathbf{r}_0) &= 0\\ \langle 2,-2,1\rangle\cdot\langle x+3,y+4,z-5\rangle &= 0\\ 2(x+3)+(-2)(y+4)+1(z-5) &= 0\\ 2x-2y+z &= 7\end{aligned}$$

Example 29. Find an equation of the plane that passes through the origin and is parallel to the plane $2x-y+3z=1$.

In this instance, we have $\mathbf{r}_0 = \langle 0,0,0\rangle$ and $\mathbf{n} = \langle 2,-1,3\rangle$ [a]. Therefore, the equation for the plane is

$$\begin{aligned}\mathbf{n}\cdot(\mathbf{r}-\mathbf{r}_0) &= 0\\ \langle 2,-1,3\rangle\cdot(\langle x,y,z\rangle-\langle 0,0,0\rangle) &= 0\\ \langle 2,-1,3\rangle\cdot\langle x,y,z\rangle &= 0\\ 2x-y+3z &= 0\end{aligned}$$

[a] Which is the normal vector of the second plane but the two planes are parallel..

Example 30. Find the point where the given lines intersect:

$$\mathbf{r}_1 = \langle 1,1,0\rangle + t\langle 1,-1,2\rangle,$$

and

$$\mathbf{r}_2 = \langle 2,0,2\rangle + s\langle -1,1,0\rangle.$$

We can equate the two lines

$$\begin{aligned}1+t &= 2-s\\ 1-t &= s\\ 2t &= 2.\end{aligned}$$

Example 31. Solving this, we obtain $t=1, s=0$ and the point of intersection* $(2,0,2)$. Find an equation of the plane that contains these lines. From the above, we have

$$\mathbf{v}_1 = \langle 1,-1,2\rangle, \quad \mathbf{v}_2 = \langle -1,1,0\rangle.$$

We can find the normal vector by taking the cross product of these vectors to form a vector perpendicular to the plane.

$$\begin{aligned}\mathbf{n} &= \mathbf{v}_1\times\mathbf{v}_2\\ &= \langle -2,-2,0\rangle.\end{aligned}$$

Thus, the equation of the plane is

$$\begin{aligned}\mathbf{n}\cdot(\mathbf{r}-\mathbf{r}_0) &= \langle -2,-2,0\rangle\cdot\langle x-2,y-0,z-2\rangle\\ -2x-2y+4 &= 0\\ -x-y+2 &= 0\end{aligned}$$

Note that if two lines on a plane are given, we can also find the equation of the plane by taking their cross product of those lines.

Theorem 7 (Shortest Distance From a Point to a Plane). *The (shortest) distance D between a point (x_1, y_1, z_1) and the plane $ax + by + cz - d = 0$ is*

$$D = \frac{|ax_1 + by_1 + cz_1 - d|}{\sqrt{a^2 + b^2 + c^2}}$$

Proof. Let $R = (x_0, y_0, z_0)$ be a point on the plane and $E = (x_1, y_1, z_1)$ be the point given. We can can form a vector between these two points

$$\mathbf{v} = E - R.$$

We have that

$$\begin{aligned}
\|\mathbf{v} \cdot \mathbf{n}\| &= \|\mathbf{v}\| \|\mathbf{n}\| \cos\theta \\
&= \|\mathbf{n}\| D && \text{(The projection of the vector)} \\
D &= \frac{\|\mathbf{v} \cdot \mathbf{n}\|}{\|\mathbf{n}\|} \\
&= \frac{\|\langle x_1 - x_0, y_1 - y_0, z_1 - z_0 \rangle \cdot \langle a, b, c \rangle\|}{\|\langle a, b, c \rangle\|} \\
&= \frac{|a(x_1 - x_0) + b(y_1 - y_0) + c(z_1 - z_0)|}{\sqrt{a^2 + b^2 + c^2}} \\
&= \frac{|ax_1 + by_1 + cz_1 - (ax_0 + by_0 + cy_0)|}{\sqrt{a^2 + b^2 + c^2}} \\
&= \frac{|ax_1 + by_1 + cz_1 - d|}{\sqrt{a^2 + b^2 + c^2}}
\end{aligned}$$

□

Example 32. Find the equation of the plane that passes through the point $P = (1, 2, 3)$ and contains the line

$$x = 2 + 4t, \quad y = 1 + 5t, \quad z = -1 + 3t.$$

We have one vector already, however, we need another one. We can construct one from P and some point on the line given, then take their cross product to find a normal vector. We can find a point on the line we already have by plugging in some value of t. Let $t = 0$, then

$$Q = \langle x, y, z \rangle = \langle 2, 1. - 1 \rangle.$$

Then we can construct the vector

$$\overrightarrow{PQ} = P - Q = \langle 1, -1, -4 \rangle.$$

Now we find the normal vector

$$\begin{aligned}
\mathbf{n} &= \overrightarrow{PQ} \times \mathbf{v} \\
&= \langle 1, -1, -4 \rangle \times \langle 4, 5, 3 \rangle \\
&= \langle 17, -19, 9 \rangle.
\end{aligned}$$

For the equation of the plane, we have

$$17x - 19y + 9z = \underbrace{17(1) - 19(2) + 9(3)}_{\text{some point on the plane}}$$

$$17x - 19y + yz = 6$$

Quadric Surfaces

Definition 21 (Elliptic Paraboloid). An elliptic paraboloid has the equation

$$z = x^2 + y^2.$$

Taking its cross section* can give you either a paraboloid or an ellipse, hence the name.

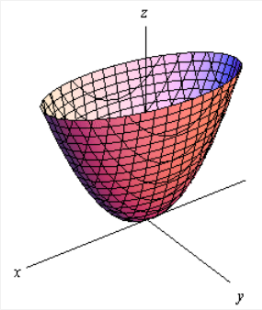


Figure 2.5: An elliptic paraboloid

The term which is not squared and its sign determines the axis and direction in which the paraboloid is open. In the case of the equation above, our paraboloid opens in the upward z-direction.

Definition 22 (Ellipsoids). In two dimensions, ellipsoids have the equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

In three dimensional space, ellipsoids have the equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

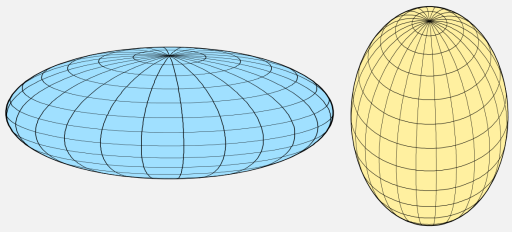


Figure 2.6: Ellipsoids

No matter how you take a cross section*, you will always get an ellipse. If a, b, c are the same, then we have

$$x^2 + y^2 + z^2 = a^2,$$

which is the equation for a sphere.

Definition 23 (Cone). A 3-dimensional cone has the equation

$$z^2 = x^2 + y^2.$$

A cone will have two parts, as shown in the figure below.

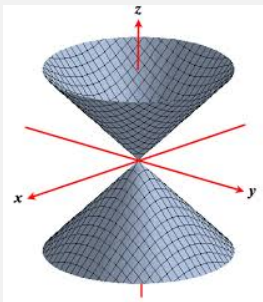


Figure 2.7: A cone.

It has the general form

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}.$$

Once again, the term that is set equal to the other terms determines the direction that the cone opens.

Definition 24 (Hyperboloids of One Sheet). Hyperboloids of one sheet have the following equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{y^2}{c^2} = 1.$$

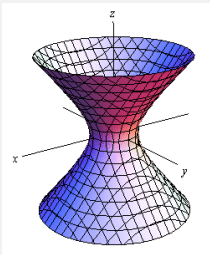


Figure 2.8: A hyperboloid of one sheet.

Definition 25 (Hyperboloids of Two Sheets). Hyperboloids of two sheets have the equation

$$\frac{z^2}{c^2} - \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

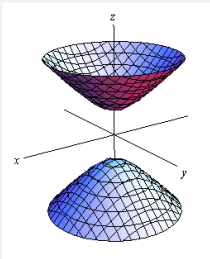


Figure 2.9: A hyperboloid of two sheets.

Definition 26 (Hyperbolic Paraboloid). A hyperbolic paraboloid has the equation

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{z}{c}.$$

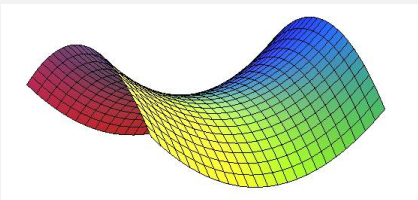


Figure 2.10: A hyperbolid paraboloid.

Cutting this in one direction, you obtain a hyperboloid and in another you obtain a paraboloid.

Example 33.

1. Sketch $\frac{x^2}{9} + \frac{y^2}{16} + \frac{z^2}{25}$.

We can begin by identifying intercepts. When x and y are zero,

$$z = \pm 5.$$

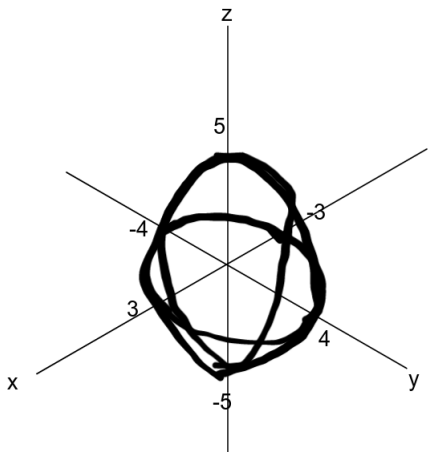
When x and z are zero,

$$y = \pm 4.$$

When y and z are zero,

$$x = \pm 3.$$

Taking the cross section in every direction, we obtain an ellipsoid.



2. Sketch $x^2 - \frac{y^2}{9} + z^2 = 1$.

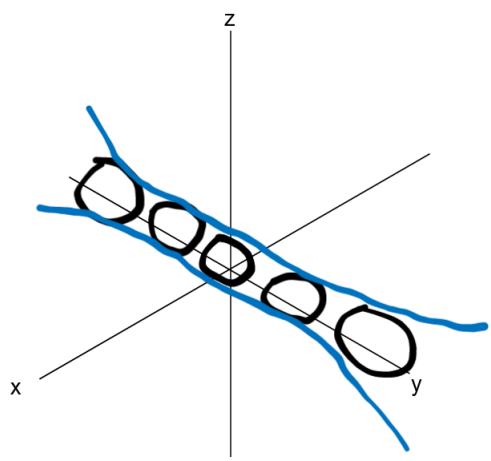
   When $y = 0$, we have

$$x^2 + z^2 = 1,$$

   or a circle of radius 1. When $y = \pm 1$, we have another circle

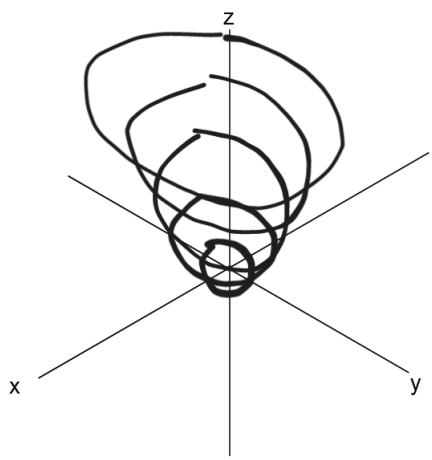
$$x^2 - \frac{1}{9} + z^2 = 1 \Rightarrow x^2 + z^2 = 1\frac{1}{9}.$$

   Continuing this, we obtain a hyperboloid of one sheet.



3. Sketch $z = \frac{x^2}{4} + \frac{y^2}{9}$.

   Again, we can try plugging in different values of z and obtain a ellipsoid.



Chapter 3

Vector Valued Functions

Vector Functions and Space Curves

Curves in a three-dimensional space have the following form

$$\begin{aligned}\mathbf{r} &= \mathbf{r}(t)\\ &= x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}\\ &= \langle x(t), y(t), z(t)\rangle,\end{aligned}$$

where $t \in \mathbb{R}$. Most curves are very hard to draw or visualize, so we will consider only simple ones. For example

$$\begin{aligned}\mathbf{r}(t) &= \langle -1, 2, 6\rangle + t\langle -5, 2, 1\rangle\\ &= (-1-5t)\mathbf{i} + (2+2t)\mathbf{j} + (6+t)\mathbf{k}.\end{aligned}$$

Alternatively, we could have a vector function in 2-dimensions

$$\mathbf{r}(t) = \mathbf{i}\cos t + \mathbf{j}\sin t = \langle \cos t, \sin t\rangle.$$

One more example is the *helix* function

$$\mathbf{r}(t) = \mathbf{i}\cos t + \mathbf{j}\sin t + \mathbf{k}t.$$

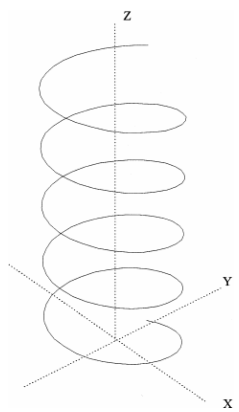


Figure 3.1: The helix function.

Construction of a line segment between two points

In order to construct a line segment between two points, we can use the parametrization

$$\mathbf{r}(t) = (1-t)\mathbf{P} + t\mathbf{Q}, \quad 0 \le t \le 1.$$

Example 34. Let $P = (1,3,-2)$ and $Q = (2,-1,3)$. Find the vector and parametric equations of the line segment that joins P and Q. We have the vector equation

$$\begin{aligned}
\mathbf{r}(t) &= (1-t)\mathbf{P} + t\mathbf{Q} \\
&= (1-t)\langle 1,3,-2\rangle + t\langle 2,-1,3\rangle \\
&= \langle 1-t, 3-3t, -2+2t\rangle + t\langle 2,-1,3\rangle \\
&= \langle 1,3,-2\rangle + t\langle -1+2, -3-1, 2+3\rangle \\
&= \langle 1,3,-2\rangle + t\langle -1,-4,5\rangle.
\end{aligned}$$

Furthermore, the parametric equations are

$$x = 1+t, \quad y = 3-4t, \quad z = -2+5t.$$

Calculus of Vector Functions

Definition 27. *The derivative of a vector valued function* $\mathbf{r}(t)$ *is defined as*

$$\mathbf{r}'(t) = \lim_{h\to 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}$$

Another interpretation is that $\mathbf{r}'(t)$ is a tangent vector of the curve.

Theorem 8. *Let* $\mathbf{r}(t) = \langle x(t), y(t), z(t)\rangle$*, where* $x(t), y(t), z(t)$ *are differentiable functions. Then*

$$\mathbf{r}'(t) = \langle x'(t), y'(t), z'(t)\rangle.$$

In this case, $\mathbf{r}(t)$ is always perpendicular to $\mathbf{r}'(t)$.

Theorem 9. *If* $\|\mathbf{r}(t)\| = c$*, where* c *is a constant, then*

$$\mathbf{r}(t)\cdot\mathbf{r}'(t)$$

Proof. Assume that $\|\mathbf{r}(t)\| = c$.

$$\begin{aligned}
c^2 &= \|\mathbf{r}(t)\|^2 \\
&= \mathbf{r}(t)\cdot\mathbf{r}(t). && (\|\mathbf{v}\|^2 = \mathbf{v}\cdot\mathbf{v})
\end{aligned}$$

Differentiating both sides with respect to t, we obtain

$$\begin{aligned}
(c^2)' &= (\mathbf{r}(t)\cdot\mathbf{r}(t))' \\
0 &= \mathbf{r}(t)\cdot\mathbf{r}'(t) + \mathbf{r}'(t)\cdot\mathbf{r}(t) \\
0 &= 2\mathbf{r}(t)\cdot r'(t) \\
0 &= \mathbf{r}(t)\cdot\mathbf{r}'(t)
\end{aligned}$$

□

Definition 28 (Unit Tangent Line). The unit tangent line is defined as

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|},$$

where $\|\mathbf{r}'(t)\| = \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2}$.

Theorem 10 (Properties of Vector Derivatives). Some properties of derivatives of vector valued functions include

1. $\frac{d}{dt}[\mathbf{u} \cdot \mathbf{v}] = \mathbf{u}' \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{v}'$
2. $\frac{d}{dt}[\mathbf{u} \times \mathbf{v}] = \mathbf{u}' \times \mathbf{v} + \mathbf{u} \times \mathbf{v}'$

Example 35.

1. Investigate the derivative of $\mathbf{r}(t) = \langle \cos t, \sin t \rangle$.

$$\mathbf{r}'(t) = \langle -\sin t, \cos t \rangle.$$

2. Find the derivative of $\mathbf{r}(t) = (1 + t^3)\mathbf{i} + (te^{-t})\mathbf{j} + \sin 2t\mathbf{k}$.

$$\mathbf{r}'(t) = 3t^2\mathbf{i} + (t(-e^{-t}) + e^{-t})\mathbf{j} + 2\cos 2t\mathbf{k}.$$

Theorem 11. *Let $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$, where $x(t), y(t), z(t)$ are integratable functions. Then*

$$\int_a^b \mathbf{r}(t)\,dt = \left\langle \int_a^b x(t)\,dt, \int_a^b y(t)\,dt, \int_a^b z(t)\,dt \right\rangle$$

Example 36.

1. *Find the integral of* $\mathbf{r}(t) = 2\cos t\mathbf{i} + \sin^2 t\mathbf{j} + 2t\mathbf{k}$.

$$\begin{aligned} \int \mathbf{r}(t)\,dt &= \int \left\langle 2\cos t, \sin^2 t, 2t \right\rangle dt \\ &= \langle 2\sin t, -\frac{1}{4}\sin 2t, t^2 \rangle + C. \end{aligned}$$

Note that $C \in \mathbb{R}^3$, i.e. $C = \langle c_1, c_2, c_3 \rangle$.

2. *Find the integral of* $\int_0^{\frac{\pi}{2}} \langle \cos 2t, \sin 2t \rangle\,dt$.

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \langle \cos 2t, \sin 2t \rangle\,dt &= \left\langle \frac{1}{2}\sin 2t, -\frac{1}{2}\cos 2t \right\rangle \Big|_{t=0}^{t=\frac{\pi}{2}} \\ &= \left\langle \frac{\sin\pi - \sin 0}{2}, \frac{-\cos\pi + \cos 0}{2} \right\rangle \\ &= \langle 0, 1 \rangle. \end{aligned}$$

3. *Given* $\mathbf{r}(t) = \left\langle t\sqrt{t+5}, t^2+2, \frac{e^t-1}{t} \right\rangle$, *find*

   (a) *Domain of* $\mathbf{r}(t)$.

   We need $t \geq -5$ to satisfy the non-negativity of the square root and $t \neq 0$ because it is in the denominator of the third term. So we have the domain

$$[-5,0) \cup (0,\infty).$$

   (b) *Determine* $\{t \mid \mathbf{r}(t) \text{ is continuous}\}$.

   We can just choose a continuous chunk from the domain defined above, e.g. $[1,10]$.

   (c) *Compute* $\lim_{t\to 0} \mathbf{r}(t)$

$$\begin{aligned}
\lim_{t\to 0} t\sqrt{t+5} &= 0 \\
\lim_{t\to 0} (t^2+2) &= 2 \\
\lim_{t\to 0} \frac{e^t-1}{t} &= \lim_{t\to 0} e^t && \text{(L'Hospital's)} \\
&= 1
\end{aligned}$$

   So the limit is

$$\lim_{t\to 0} \mathbf{r}(t) = \langle 0,2,1 \rangle.$$

4. Sketch $\mathbf{r}(t) = \langle t, t^2, 0 \rangle$.

We see that

$$y = t^2 = (t)^2 = x^2,$$

so we have a parabola that opens in the positive y-direction..

Example 37. Find the velocity and position vector of an object where the acceleration and initial velocity is given.

$$\begin{aligned}
\mathbf{a}(t) &= 2\mathbf{i} + 2t\mathbf{k} \\
\mathbf{v}(0) &= 3\mathbf{i} - \mathbf{j} \\
\mathbf{r}(0) &= \mathbf{j} + \mathbf{k},
\end{aligned}$$

where $\mathbf{r}$ is our position vector, velocity $\mathbf{v}(t)$ is $\mathbf{r}'(t)$ and acceleration is $\mathbf{v}'(t)$.

To do so, we first integrate $\mathbf{a}(t)$

$$\begin{aligned}
\mathbf{a}(t) &= 2\mathbf{i} + 2t\mathbf{k} \\
\int \mathbf{a}(t)\,dt &= \int \langle 2,0,2t \rangle\,dt \\
&= \langle 2t, 0, t^2 \rangle + C.
\end{aligned}$$

As we have $\mathbf{v}(t) = \int \mathbf{a}(t)\,dt$, it must be the case that

$$\mathbf{v}(0) = 3\mathbf{i} - \mathbf{j} = C.$$

So our velocity vector $\mathbf{v}(t)$ is

$$\mathbf{v}(t) = \langle 2t, 0, t^2 \rangle + \langle 3, -1, 0 \rangle = \langle 2t+3, -1, t^2 \rangle.$$

To determine our position vector

$$\begin{aligned}\mathbf{r}(t) &= \int \mathbf{v}(t)\,dt \\ &= \int \langle 2t+3, -1, t^2 \rangle\,dt \\ &= \left\langle t^2+3t, -t, \frac{t^3}{3} \right\rangle + C_2\end{aligned}$$

To determine C_2, we use the initial condition that $\mathbf{r}(0) = \mathbf{j} + \mathbf{k}$. Thus, we have

$$\mathbf{j} + \mathbf{k} = \mathbf{r}(0) = C_2.$$

So

$$\begin{aligned}\mathbf{r}(t) &= \left\langle t^2+3t, -t, \frac{t^3}{3} \right\rangle + \mathbf{j} + \mathbf{k} \\ &= \left\langle t^2+3t, 1-t, 1+\frac{t^3}{3} \right\rangle.\end{aligned}$$

Arc Length

Definition 29 (Arc Length). The arc-length L for a 2d-space $\mathbf{r}(t) = \langle x(t), y(t) \rangle, \quad a \le t \le b$ is

$$L = \lim_{n\to\infty} \sum_{i=1}^{n} \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2} = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt.$$

For a 3d-space $\mathbf{r}(t) = \langle x(t), y(t), z(t), \quad a \le t \le b$

$$L = \lim_{n\to\infty} \sum_{i=1}^{n} \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2 + (\Delta z_i)^2} = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\,dt.$$

In other terms, we have $L = \int_a^b \|\mathbf{r}'(t)\|\,dt$.

Example 38. Find the length of the arc of the circular helix with vector equation

$$\mathbf{r}(t) = \cos t\,\mathbf{i} + \sin t\,\mathbf{j} + t\,\mathbf{k},$$

from $(1,0,0)$ to $(1,0,2\pi)$. We see that $t \in [0, 2\pi]$. Pre-calculating the derivative, we have

$$\mathbf{r}'(t) = -\sin t\,\mathbf{i} + \cos t\,\mathbf{j} + \mathbf{k}.$$

Thus, the arc length is

$$\begin{aligned} L &= \int_0^{2\pi} \sqrt{(-\sin t)^2 + (\cos t)^2 + 1^2}\, dt \\ &= \int_0^{2\pi} \sqrt{\sin^2 t + \cos^2 t + 1}\, dt \\ &= 2\sqrt{2}\pi. \end{aligned}$$

Note that without the y-coordinate, we would have a unit circle, the circumference of which would be only 2π.

Example 39. Find the length of the arc of vector equation $\mathbf{r}(t) = \mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k}$ from $(1,0,0)$ to $(1,1,1)$. First, we must determine a and b. We can do so by plugging both our points in

$$\begin{aligned} \mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k} = (1,0,0) &\implies t = 0 \\ \mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k} = (1,1,1) &\implies t = 1. \end{aligned}$$

So we have $t \in [0,1]$. We have

$$\mathbf{r}'(t) = \langle 0, 2t 3t^3 \rangle.$$

So the arc length is

$$\begin{aligned} L &= \int_0^1 \|\mathbf{r}'(t)\|\, dt \\ &= \int_0^1 \sqrt{0^2 + (2t)^2 + (3t^2)^2}\, dt \\ &= \int_0^1 \sqrt{4t^2 + 9t^2}\, dt \\ &= \int_0^1 t\sqrt{4 + 9t^2}\, dt. \end{aligned}$$

Let $u = 4 + 9t^2, du = 18t\, dt, \frac{1}{18}\, du = t\, dt$. Thus

$$\begin{aligned} L &= \int_0^1 t\sqrt{4 + 9t^2}\, dt \\ &= \frac{1}{18}\int_0^1 \sqrt{u}\, du \\ &= \frac{2}{3}\cdot\frac{1}{18}\cdot u^{3/2} + C\Big|_0^1 \\ &= \frac{1}{27}u^{3/2} + C\Big|_0^1 \\ &= \frac{1}{27}(4 + 9t)^{3/2} + C\Big|_0^1 \\ &= \frac{1}{27}(13^{3/2} - 8) \end{aligned}$$

We can also define arc length as a function which is called the *arc length function*

$$s(t) = \int_a^t \|\mathbf{r}'(u)\|\, du.$$

The constant a is called the reference point. Equivalently, we have

$$\frac{ds}{dt} = \|\mathbf{r}'(t)\|.$$

Example 40. For a circular helix, $\mathbf{r}(t) = \langle \cos t, \sin t, t\rangle$, we have $s(t) = \sqrt{2}t$. Similarly, if we change a variable in the circular helix,

$$\mathbf{r}(s) = \left\langle \cos\left(\frac{s}{\sqrt{2}}\right), \sin\left(\frac{s}{\sqrt{2}}\right), \frac{s}{\sqrt{2}}\right\rangle.$$

We note that $\left\|\frac{d\mathbf{r}}{ds}\right\| = 1$. This is done by plugging $t(s)$ into $\mathbf{r}$.

$$\begin{aligned}\mathbf{r}(t(s)) &= \langle \cos t(s), \sin t(s), t(s)\rangle \\ &= \left\langle \cos\left(\frac{s}{\sqrt{2}}\right), \sin\left(\frac{s}{\sqrt{2}}\right), \frac{s}{\sqrt{2}}\right\rangle.\end{aligned}$$

With this information, we can determine where we are on the curve after traveling some distance. For example, we can calculate where we are on the curve after traveling for a distance of $\sqrt{2}\pi$. We plug in $s = \sqrt{2}\pi$

$$\begin{aligned}\left\langle \cos\left(\frac{\sqrt{2}\pi}{\sqrt{2}}\right), \sin\left(\frac{\sqrt{2}\pi}{\sqrt{2}}\right), \frac{\sqrt{2}\pi}{\sqrt{2}}\right\rangle &= \langle \cos\pi, \sin\pi, \pi\rangle \\ &= \langle -1, 0, \pi\rangle.\end{aligned}$$

Example 41. Find the arc length function for

$$\mathbf{r}(t) = \langle e^t, e^t \sin t, e^c \cos t\rangle$$

from $(1,0,1)$. We can first take the derivative

$$\mathbf{r}'(t) = \langle e^t, e^t \sin t + e^t \cos t, e^t \cos t - e^t \sin t\rangle.$$

Next, we find the norm, as $s(t) = \int_a^b \|\mathbf{r}'(u)\|\, du$.

$$\begin{aligned}\|\mathbf{r}'(t)\| &= \sqrt{(e^t)^2 + (e^t \sin t + e^t \cos t)^2 + (e^t \cos t - e^t \sin t)^2} \\ &= \sqrt{e^{2t} + e^{2t}\sin^2 t + \cancel{2e^t \sin t \cos t} + e^{2t}\cos^2 t + e^{2t}\cos^2 t - \cancel{2e^t \sin t \cos t} + e^{2t}\sin^2 t} \\ &= \sqrt{e^{2t}\left(1 + 2\sin^2 t + 2\cos^2 t\right)} \\ &= \sqrt{e^{2t}(1 + 2(1))} \\ &= e^t\sqrt{3}\end{aligned}$$

So the arc length function is

$$\begin{aligned}s(t) &= \int_0^t \sqrt{3}e^u\, du \\ &= \sqrt{3}e^u\Big|_{u=0}^{u=t} \\ &= \sqrt{3}\left(e^t - e^0\right) \\ &= \sqrt{3}e^t - \sqrt{3}.\end{aligned}$$

Chapter 4

Partial Derivatives

Functions of Two or More Variables

Let $z = f(x, y)$ for (x, y) in the *domain* D. The set of values that $f(x, y)$ takes on is called *range*.

Example 42. Find the domain of the following functions and evaluate $f(2, 1)$.

a) $f(x, y) = \frac{\sqrt{x+y+1}}{x-1}$.

b) $f(x, y) = \sqrt{9 - x^2 - y^2}$.

For (a), we require $x + y + 1 \geq 0$ and $x - 1 \neq 0$. In other terms, the domain is

$$\{(x, y) \in \mathbb{R}^2 \mid x + y + 1 \geq 0 \text{ and } x \neq 1\}.$$

For (b), we again require the term inside the root to be non-negative. Thus, the domain is

$$\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 9\},$$

which is a disk of radius 3.

The graph of $f(x, y)$ is a 2-dimensional surface in 3d-space. Normally it's not that easy to draw these surfaces.

Example 43. Sketch the graph of the function in part (b) and find its range.

We have that

$$\begin{aligned} z &= \sqrt{9 - x^2 - y^2} \\ z^2 &= 9 - x^2 - y^2 \\ x^2 + y^2 + z^2 &= 9, \end{aligned}$$

which is a sphere of radius 9.

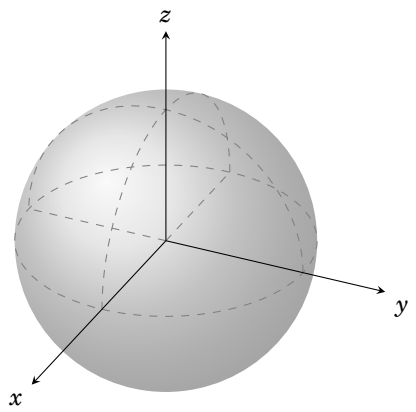


Figure 4.1: A sphere of radius 3

Its range is

$$R_f = \{z \mid 0 \le z \le 3\}.$$

Example 44. Given $z = \frac{\sqrt{y-x^2}}{1-x^2}$, find and sketch the domain.

We require that $1-x^2 \neq 0$ and $y-x^2 \geq 0$. In other terms, we want $x \neq \pm 1$ and $y \geq x^2$.

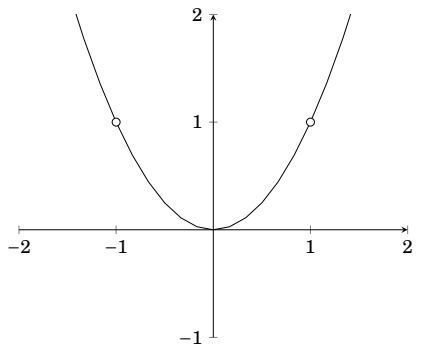


Figure 4.2: The domain of $z = \frac{\sqrt{y-x^2}}{1-x^2}$

Example 45. Find the domain of $f(x,y) = \sin^{-1}(x+y)$.

Its domain is

$$D = \{(x,y) \in \mathbb{R}^2 \mid -1 \le x+y \le 1\}.$$

Limits and Continuity

We first consider the limit of a function of one variable $f(x)$ is written as $\lim_{x\to a} f(x) = L$. We only consider the limit by approaching from two directions, left or right. Limits of functions of two

variables are analogous to those of one

$$\lim_{(x,y)\to(a,b)} f(x,y) = L.$$

Unlike one variable, the limit can be approached from infinitely many directions. The goal is to show whether the limit exists or does not exist.

Example 46. Compute the limit

$$\lim_{(x,y)\to(1,2)} \sqrt{x^3+3+6y}.$$

We have that

$$\lim_{(x,y)\to(1,2)} \sqrt{x^3+3+6y} = \sqrt{(1)^3+3+6(2)} = \sqrt{16} = 4.$$

Example 47. Show that the limit below does not exist

$$\lim_{(x,y)\to(0,0)} \frac{x^2-y^2}{x^2+y^2}.$$

We have that when $y=0$

$$\begin{aligned}\lim_{(x,y)\to(0,0)} \frac{x^2-y^2}{x^2+y^2} &= \lim_{(x,0)\to(0,0)} \frac{x^2-0}{x^2+0^2}\\ &= \lim_{x\to 0} \frac{x^2}{x^2}\\ &= 1.\end{aligned}$$

When $x=0$, we have

$$\begin{aligned}\lim_{(x,y)\to(0,0)} \frac{x^2-y^2}{x^2+y^2} &= \lim_{(0,y)\to(0,0)} \frac{0^2-y^2}{0^2+y^2}\\ &= \lim_{y\to 0} \frac{-y^2}{y^2}\\ &= -1.\end{aligned}$$

Since the two limits approaching from different directions are not equal, then $\lim_{(x,y)\to(0,0)} \frac{x^2-y^2}{x^2+y^2}$ does not exist.

Example 48. Show that the limit does not exist

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}.$$

We can use the method we did for the previous example but both of the limits would be the same: zero. This does not imply that the limit would be zero however. Instead we can try a

different method based on the form of the function. Along the line $y = x$, we have that

$$\begin{aligned}\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} &= \lim_{(x,x)\to(0,0)} \frac{x^2}{2x^2} \\ &= \lim_{x\to 0} \frac{x^2}{2x^2} \\ &= \frac{1}{2}.\end{aligned}$$

Since this is not equal to zero, that means the limit does not exist.

Example 49. Show that the limit does not exist

$$\lim_{(x,y)\to(0,0)} \frac{xy^2}{x^2+y^4}.$$

Consider when $x = 0$, then

$$\lim_{(0,y)\to(0,0)} \frac{(0)(y^2)}{(0)+y^4} = 0.$$

When $x = y^2$ then

$$\lim_{(y^2,y)\to(0,0)} \frac{y^4}{2y^4} = \frac{1}{2}.$$

So the limit does not exist.

For more difficult cases, it is much more involved to show that the limit exists. We have to apply the ϵ-δ argument. For some cases we can use parametrization to simplify things

Example 50. Compute the limit

$$\lim_{(x,y)\to(0,0)} \frac{3x^2y}{x^2+y^2}.$$

Recall that

$$\begin{aligned}x^2+y^2 &= r^2 \\ &= r^2(\cos^2\theta + \sin^2\theta) \\ &= (r\cos\theta)^2 + (r\sin\cos\theta)^2.\end{aligned}$$

So we let $x = r\cos\theta$ and $y = r\sin\theta$.

$$\begin{aligned}\lim_{(x,y)\to(0,0)} \frac{3x^2y}{x^2+y^2} &= \lim_{r\to 0} \frac{3(r\cos\theta)^2(r\sin\theta)}{r^2} \\ &= \lim_{r\to 0} 3r\cos^2\theta\sin\theta \\ &= 0.\end{aligned}$$

Note that this works because we know $\cos^2\theta\sin\theta$ will give us a finite value, as both functions are bounded (and thus cannnot be infinite, giving us an indeterminate value).

Definition 30 (Continuity). A function f of two variables is called continuous at the point (a,b) if

$$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b).$$

To verify continuity, we check three conditions

1. f is defined at $(x,y) = (a,b)$.
2. $\lim_{(x,y)\to(a,b)} f(x,y)$ exists.
3. $f(a,b) = \lim_{(x,y)\to(a,b)} f(x,y)$.

Example 51. Where is the function $g(x,y) = \arctan\left(\frac{y}{x}\right)$ continuous?

Recall first, the graph for arctan

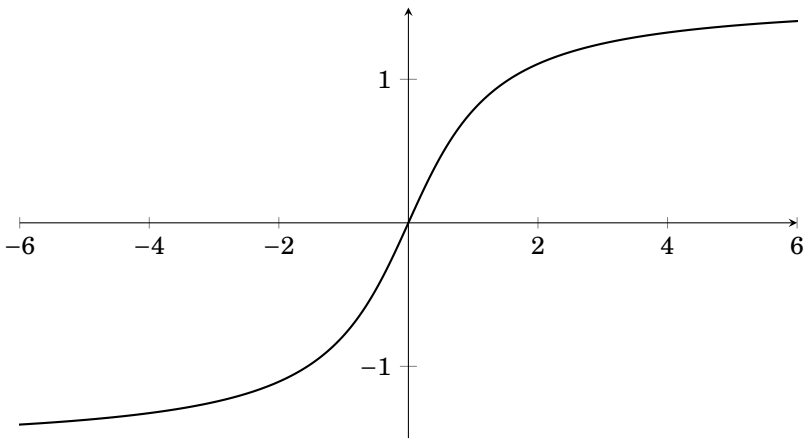


Figure 4.3: The arctangent function

Its domain is all real numbers excluding $x = 0$.

$$D_f = \{(x,y) \in \mathbb{R} \mid x \neq 0\}.$$

Example 52. Where is the function $f(x,y) = \frac{x^2-y^2}{x^2+y^2}$ continuous?

Since the limit as $(x,y) \to (0,0)$ does not exist, f is not continuous at $(0,0)$. So, f is continuous everywhere except $(0,0)$.

$$D_f = \mathbb{R}^2 \setminus \{(0,0)\}.$$

Example 53. Is the function below continuous at $(0,0)$?

$$f(x,y) = \begin{cases} \frac{x^4-y^4}{x^2+y^2}, & \text{if } (x,y) \neq (0,0) \\ 0, & \text{if } (x,y) = (0,0) \end{cases}.$$

Consulting the definition of continuity, we see that

1. f is defined at $(0,0)$ and equal to zero.
2. The limit of f as $(x,y) \to (0,0)$ exists and is equal to zero.
3. We also see that $f(0,0) = \lim_{(x,y)\to(0,0)} f(x,y)$.

Thus, f is continuous at $(0,0)$.

Definition 31 (Traces). Traces (or level curves) of a function of two variables are curves with equation $f(x,y) = k$.

Example 54. Sketch level curves of $f(x,y) = x^2 + y^2$.

We can sketch the level curves by setting $f(x,y) = z$ to some constant.

$$\begin{aligned} z = 0 &\implies x^2 + y^2 = 0 \\ z = 1 &\implies x^2 + y^2 = 1 \\ z = 2 &\implies x^2 + y^2 = 2 \\ &\vdots \end{aligned}$$

Partial Derivatives

Definition 32. Partial derivatives with respect to x and y are the following

$$\begin{aligned} f_x(a,b) &= \lim_{h\to 0} \frac{f(a+h,b) - f(a,b)}{h} \\ f_y(a,b) &= \lim_{h\to 0} \frac{f(a,b+h) - f(a,b)}{h}. \end{aligned}$$

In terms of notation

$$f_x = f_x(x,y) = \frac{\partial f}{\partial x} = \frac{\partial f(x,y)}{\partial x} = \frac{\partial z}{\partial x}.$$

Example 55. If $f(x,y) = x^3 + x^2y^3 - 2y^2$, find $f_x(2,1)$ and $f_y(2,1)$.

$$\begin{aligned} f_x &= 3x^2 + 2xy^3 \\ f_x(2,1) &= 3(2)^2 + 2(2)(1)^3 \\ &= 16 \\ f_y &= 3x^2y^2 - 4y \\ f_y(2,1) &= 3(2)^2(1)^2 - 4(1) \\ &= 8. \end{aligned}$$

Example 56. If $f(x,y) = \sin\left(\frac{x}{1+y}\right)$, calculate $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

$$\begin{aligned}
\frac{\partial f}{\partial x} &= \left[\cos\left(\frac{x}{1+y}\right)\right] \cdot \frac{\partial}{\partial x}\left(\frac{x}{1+y}\right) \\
&= \frac{1}{1+y} \cdot \cos\left(\frac{x}{1+y}\right) \\
\frac{\partial f}{\partial y} &= \left[\cos\left(\frac{x}{1+y}\right)\right] \cdot \frac{\partial}{\partial y}\left(\frac{x}{1+y}\right) \\
&= \left[\cos\left(\frac{x}{1+y}\right)\right]\left(-\frac{x}{(1+y)^2}\right).
\end{aligned}$$

Partial Derivatives Viewed as Rates of Change and Slopes

The partial derivative with respect to x, $f_x(x_0, y_0)$, is the slope of the surface in the x-direction at (x_0, y_0). Similarly, $f_y(x_0, y_0)$ is the slope of the surface in the y-direction at (x_0, y_0).

Example 57. Let $f(x,y) = x^2y + 5y^3$.

(a) Find the slope of the surface $z = f(x,y)$ in the x-direction at $(1,-2)$.

(b) Find the slope of the surface $z = f(x,y)$ in the y-direction at $(1,-2)$.

For part a, we have that

$$f_x = 2xy \implies f_x(1,-2) = 2(1)(-2) = -4.$$

For part b, we have that

$$f_y = x^2 + 15y^2 \implies f_y(1,-2) = (1)^2 + 15(-2)^2 = 61.$$

Higher Derivatives

We can apply partial derivative more than single time. Consider the second partial derivatives

$$(f_x)_x, \quad (f_x)_y, \quad (f_y)_x, \quad (f_y)_y.$$

Example 58. Find the second partial derivatives of

$$f(x,y) = x^3 + x^2y^3 - 2y^2.$$

We have that the first partial derivatives are

$$\begin{aligned}
f_x &= 3x^2 + 2xy^3 \\
f_y &= 3x^2y^2 - 4y.
\end{aligned}$$

Next we find the second partial derivatives

$$\begin{aligned}
f_{xx} &= \frac{\partial}{\partial x}(f_x) \\
&= \frac{\partial}{\partial x}(3x^2 + 2xy^3) \\
&= 6x + 2y^3.
\end{aligned}$$

$$\begin{aligned} f_{xy} &= \frac{\partial}{\partial y}(f_x) \\ &= \frac{\partial}{\partial y}(3x^2 + 2xy^3) \\ &= 6xy^2. \end{aligned}$$

$$\begin{aligned} f_{yy} &= \frac{\partial}{\partial y}(f_y) \\ &= \frac{\partial}{\partial y}(3x^2y^2 - 4y) \\ &= 6x^2y - 4. \end{aligned}$$

$$\begin{aligned} f_{yx} &= \frac{\partial}{\partial x} f_y \\ &= \frac{\partial}{\partial x}(3x^2y^2 - 4y) \\ &= 6xy^2. \end{aligned}$$

Theorem 12 (Clairaut's Theorem). If the functions f_{xy} and f_{yx} are both continuous on a disk D, then

$$f_{xy}(a,b) = f_{yx}(a,b).$$

Example 59. Laplace's equation is given by

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

Verify that $u(x,y) = e^x \sin(y)$ is a solution of Laplace's equation.

$$\begin{aligned} \frac{\partial u}{\partial x} &= e^x \sin y \\ \frac{\partial u}{\partial y} &= e^x \cos y \\ \frac{\partial^2 u}{\partial x^2} &= e^x \sin y \\ \frac{\partial^2 u}{\partial y^2} &= -e^x \sin y. \end{aligned}$$

Thus, we have that

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = e^x \sin y - e^x \sin y = 0.$$

Thus, $u(x,y) = e^x \sin(y)$ is a solution of Laplace's equation.

Implicit Differentiation

The implicit differentiation of functions of two variables can be done through the partial derivative and chain rule.

Example 60. Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ where z is defined implicitly as a function of x and y by the equation

$$x^3 + y^3 + z^3 + 6xyz = 1.$$

We have that for $\frac{\partial z}{\partial x}$

$$\begin{aligned}
\frac{\partial}{\partial x}\left(x^3 + y^3 + z^3 + 6xyz\right) &= \frac{\partial}{\partial x}(1) \\
3x^2 + 3z^2 \cdot \frac{\partial z}{\partial x} + 6y\left(x\frac{\partial z}{\partial x} + z\frac{\partial x}{\partial x}\right) &= 0 \\
3x^2 + 3z^2 \cdot \frac{\partial z}{\partial x} + 6xy \cdot \frac{\partial z}{\partial x} + 6yz &= 0 \\
\left(3z^2 + 6xy\right)\frac{\partial z}{\partial x} &= -3x^2 - 6yz \\
\frac{\partial z}{\partial x} &= \frac{-3x^2 - 6yz}{3z^2 + 6xy}.
\end{aligned}$$

For $\frac{\partial z}{\partial y}$, we can find it in a similar manner

$$\begin{aligned}
\frac{\partial}{\partial y}\left(x^3 + y^3 + z^3 + 6xyz\right) &= \frac{\partial}{\partial y}(1) \\
3y^2 + 3z^2 \cdot \frac{\partial z}{\partial y} + 6x\left(y\frac{\partial z}{\partial y} + z\frac{\partial y}{\partial y}\right) &= 0 \\
3y^2 + 3z^2 \cdot \frac{\partial z}{\partial y} + 6xy \cdot \frac{\partial z}{\partial y} + 6xz &= 0 \\
\left(3z^2 + 6xy\right)\frac{\partial z}{\partial y} &= -3y^2 - 6xz \\
\frac{\partial z}{\partial y} &= \frac{-3y^2 - 6xz}{3z^2 + 6xy}.
\end{aligned}$$

Thus, we have that

$$\frac{\partial z}{\partial x} = \frac{-3x^2 - 6yz}{3z^2 + 6xy}, \quad \frac{\partial z}{\partial y} = \frac{-3y^2 - 6xz}{3z^2 + 6xy}.$$

Local Linearity and Differentials

Linearization of f at (a,b) is given by

$$L(x,y) = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b).$$

We then use $L(x,y)$ to approximate $f(x,y)$.

Example 61. Find the linearization of $f(x,y) = xe^{xy}$ at $(1,0)$. Then use it to approximate

$f(1.1, -0.1)$.

$$L(x,y) = f(1,0) + f_x(1,0)(x-1) + f_y(1,0)(y-0).$$

Now that we have the linearization, we can use it to approximate $f(1.1, -0.1)$

$$\begin{aligned} f(1.1,-0.1) &\approx L(1.1,-0.1) \\ &= f(1,0) + f_x(1,0)(1.1-1) + f_y(1,0)(-0.1-0) \\ &= 1. \end{aligned}$$

Differentials

The differential of a function of one variable is

$$dy \approx \Delta y = y_1 - y_2.$$

In Calculus I, we learned that

$$d_y = f'(x)\,dx.$$

For a function of two variables, $z = f(x,y)$, the differential dz is

$$dz = f_x(x,y)\,dx + f_y(x,y)\,dy = \frac{\partial z}{\partial x}\,dx + \frac{\partial z}{\partial y}\,dy.$$

The differentials here contain the same approximation idea shown in the example above. dz is roughly $L(x,y) - f(a,b)$.

Example 62.

(a) If $z = f(x,y) = x^2 + 3xy - y^2$, find the differential dz.

(b) If x changes from 2 to 2.05 and y changes from 3 be 2.96. Find dz.

We have that for part a

$$\begin{aligned} dz &= f_x\,dx + f_y\,dy \\ &= (2x+3y)\,dx + (3x-2y)\,dy. \end{aligned}$$

For part (b), we plug in our values

$$\begin{aligned} dz &= [2(2)+3(3)](2.05-2) + [3(2)-2(3)](2.96-3) \\ &= (13)(0.05) \\ &= 0.65. \end{aligned}$$

Example 63. If $z = 5x^2 + y^2$

(a) Compute dz.

(b) Find the approximation if (x,y) changes from $(1,2)$ to $(1.05, 2.1)$.

For part (a), we have that

$$\begin{aligned} dz &= f_x\,dx + f_y\,dy \\ &= 10x\,dx + 2y\,dy. \end{aligned}$$

For part (b), we again plug in our values

$$\begin{aligned} dz &= 10(1)(1.05-1)+2(2)(2.1-2) \\ &= (10)(0.05)+4(0.1) \\ &= 0.5+0.4 \\ &= 0.9. \end{aligned}$$

Example 64. Let $f(x,y,z) = x^2\sin^3(xy+z^3)$. Find f_x.

$$\begin{aligned} \frac{\partial}{\partial x}x^2\sin^3(xy+z^3) &= x^2\left(\frac{\partial}{\partial x}\sin^3(xy+z^3)\right)+\sin^3(xy+z^3)\left(\frac{\partial}{\partial x}x^2\right) \\ &= (x^2)\left(3\sin^2(xy+z^3)\cdot\cos\left(xy+z^3\right)\cdot\frac{\partial}{\partial x}(xy+z^3)\right)+\sin^3(xy+z^3)(2x) \\ &= x^2\left(3\sin^2(xy+z^3)\cdot\cos\left(xy+z^3\right)\cdot y\right)+2x\sin^3(xy+z^3) \\ &= 3x^2y\cos\left(xy+z^3\right)\sin^2(xy+z^3)+2x\sin^3(xy+z^3). \end{aligned}$$

Example 65. Approximate $\sqrt{1.03^2+1.98^3}$.

We can find the linearization at $(1,2)$. Let $f(x,y) = \sqrt{x^2+y^3}$.

$$L(x,y) = f(1,2)+f_x(1,2)(x-1)+f_y(1,2)(y-2)$$

We know that $f(1,2) = \sqrt{(1)^2+(2)^3} = 3$. Next, we find the partial derivatives

$$\begin{aligned} f_x &= \frac{\partial}{\partial x}\sqrt{x^2+y^3} \\ &= \frac{1}{2\sqrt{x^2+y^3}}\cdot(2x) \\ &= \frac{x}{\sqrt{x^2+y^3}} \\ f_y &= \frac{\partial}{\partial y}\sqrt{x^2+y^3} \\ &= \frac{1}{2\sqrt{x^2+y^3}}\cdot(3y^2) \\ &= \frac{3y^2}{2\sqrt{x^2+y^3}}. \end{aligned}$$

Next, we plug in our points.

$$\begin{aligned} f_x(1,2) &= \frac{1}{\sqrt{9}} \\ &= \frac{1}{3} \\ f_y(1,2) &= \frac{12}{2\sqrt{9}} \\ &= 2 \end{aligned}$$

So the linearization is

$$\begin{aligned} L(x,y) &= 3 + \frac{1}{3}(1.03-1) + 2(1.98-2) \\ &\approx 2.97. \end{aligned}$$

The Chain Rule

Recall the chain rule from Calculus I, where $y = f(u)$ and $u = g(x)$.

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

Now suppose that $z = f(x,y)$ is a differentiable function of x and y, where $x = g(t)$ and $y = h(t)$ are both differentiable functions of t. Then t is a differentiable function of t and

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt}.$$

Example 66. *If $z = xy^2$, where $x = t^2$ and $y = \sqrt{t}$, find $\frac{dz}{dt}$.*

$$\begin{aligned} \frac{dz}{dt} &= \frac{\partial}{\partial x}(xy^2) \cdot \frac{d}{dt}(t^2) + \frac{\partial}{\partial y}(xy^2) \cdot \frac{d}{dt}(\sqrt{t}) \\ &= y^2 \cdot 2t + 2xy \cdot \frac{1}{2\sqrt{t}} \\ &= 2ty^2 + \frac{xy}{\sqrt{t}} \\ &= 2t(\sqrt{t})^2 + \frac{(t^2)(\sqrt{t})}{\sqrt{t}} \\ &= 2t^2 + t^2 \\ &= 3t^2. \end{aligned}$$

Another method would be to just plug in

$$z = xy^2 = (t^2)(\sqrt{t})^2 = t^3.$$

So then $\frac{dz}{dt}$ would be $3t^2$.

Example 67. If $z = xy + y^2$, where $x = \sin t$ and $y = e^{-t}$. Find $\frac{dz}{dt}$ when $t = 0$.

$$\begin{aligned} \frac{dz}{dt} &= \frac{\partial}{\partial x}(xy + y^2) \cdot \frac{d}{dt}(\sin t) + \frac{\partial}{\partial y}(xy + y^2) \cdot \frac{d}{dt}(e^{-t}) \\ &= y\cos t + (x + 2y) \cdot (-e^{-t}) \\ &= (e^{-t})\cos t - (\sin t + 2e^{-t})e^{-t} \end{aligned}$$

Then we have

$$\left.\frac{dz}{dt}\right|_{t=0} = 1 - 2 = -1.$$

In the more general case, let $z = f(x,y), x = g(s,t), y = h(s,t)$. Then

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s}$$
$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$

Example 68. *If $z = e^x \sin y$ where $x = st^2$ and $y = s^2 t$. Find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$.*

$$\begin{aligned}\frac{\partial z}{\partial s} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial s}\\ &= (e^x \sin y)\cdot(t^2) + (e^x \cos y)\cdot(2st)\\ &= e^{st^2}\sin\left(s^2 t\right)\cdot t^2 + e^{st^2}\cos\left(s^2 t\right)\cdot(2st)\\ &= e^{st^2}\left(t^2 \sin\left(s^2 t\right) + 2st\cos\left(s^2 t\right)\right)\end{aligned}$$

$$\begin{aligned}\frac{\partial z}{\partial t} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}\\ &= (e^x \sin y)\cdot(2st) + (e^x \cos y)\cdot s^2\\ &= e^x(2st\sin y + s^2\cos y)\\ &= e^{st^2}(2st\sin\left(s^2 t\right) + s^2\cos\left(s^2 t\right))\end{aligned}$$

Implicit Differentiation

One application of the chain rule is aiding in implicit differentiation. Let $F(x,y) = 0$. Then, differentiating F on both sides with respect to x using the chain rule, we obtain

$$\frac{\partial F}{\partial x}\cdot\frac{dx}{dx} + \frac{\partial F}{\partial y}\cdot\frac{dy}{dx} = 0 \implies \frac{dy}{dx} = \frac{-F_x}{F_y}.$$

Example 69. *Find $\frac{dy}{dx}$ if $x^3 + y^3 = 6xy$.*

In this instance, we construct $F(x,y) = 0$ by moving all terms to one side.

$$\begin{aligned}\frac{dy}{dx} &= \frac{-\frac{\partial}{\partial x}(x^3 + y^3 - 6xy)}{\frac{\partial}{\partial y}(x^3 + y^3 - 6xy)}\\ &= \frac{6y - 3x^2}{3y^2 - 6x}\\ &= \frac{2y - x^2}{y^2 - 2x}.\end{aligned}$$

Similarly, for $z = f(x,y)$, we let $F(x,y,z) = 0$. Then

$$\frac{\partial F}{\partial x}\cdot\frac{\partial x}{\partial x} + \frac{\partial F}{\partial y}\cdot\frac{\partial y}{\partial x} + \frac{\partial F}{\partial z}\cdot\frac{\partial z}{\partial x} = 0.$$

However, we have $\frac{\partial x}{\partial x} = 1$ and $\frac{\partial y}{\partial x} = 0$. This equation then becomes

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial z}\cdot\frac{\partial z}{\partial x} = 0 \implies \frac{\partial z}{\partial x} = \frac{-F_x}{F_z}.$$

Example 70. *Find* $\frac{\partial z}{\partial x}$ *if* $yz = \ln(x+z)$.

We have that

$$F(x,y,z) = yz - \ln(x+z).$$

Then

$$\begin{aligned}\frac{\partial z}{\partial x} &= \frac{-F_x}{F_z} \\ &= \frac{-\frac{1}{x+z}}{y - \frac{1}{x+z}} \\ &= \frac{1}{(x+z)y - 1}.\end{aligned}$$

Directional Derivatives and Gradients

Definition 33 (Gradient Vector). If f is a function of two variables, then the gradient of f is the vector function ∇f.

$$\nabla f(x,y) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle.$$

Let us illustrate this with an example.

Example 71. *Find the gradient vector of* $f(x,y) = \sin x + e^{xy}$.

$$\begin{aligned}\nabla f &= \langle f_x, f_y \rangle \\ &= \langle \cos x + ye^{xy}, xe^{xy} \rangle.\end{aligned}$$

Definition 34 (Directional Derivative). The directional derivative of f at (x_0, y_0) in the direction of a unit vector $\mathbf{u} = \langle a, b \rangle$ is

$$D_{\mathbf{u}} f(x_0, y_0) = \lim_{h \to 0} \frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h},$$

if the limit exists. Then

$$D_{\mathbf{u}} f(x,y) = f_x(x,y)a + f_y(x,y)b = \nabla f(x,y) \cdot \mathbf{u}.$$

In this instance, $D_{\mathbf{u}} f(x,y)$ is the rate of change of $f(x,y)$ in the direction $\mathbf{u}$. The partial derivatives in x and y are $D_{\langle 1,0 \rangle} f(x,y)$ and $D_{\langle 0,1 \rangle} f(x,y)$.

Example 72. *Find the directional derivative of the function* $f(x,y) = x^2y^3 - 4y$ *at the point*

$(x,y) = (2,-1)$ *in the direction of the vector* $\mathbf{v} = 2\mathbf{i} + 5\mathbf{j}$.

$$\begin{aligned} D_{\mathbf{u}} f(2,-1) &= \nabla f(2,-1) \cdot \mathbf{u} \\ &= \langle 2xy^3, 3x^2y^2 - 4 \rangle \Big|_{(x,y)=(2,-1)} \cdot \frac{\langle 2,5 \rangle}{\| \langle 2,5 \rangle \|} \\ &= \langle -4, 8 \rangle \cdot \frac{\langle 2,5 \rangle}{\sqrt{29}} \\ &= \frac{\langle -8+40 \rangle}{\sqrt{29}} \\ &= \frac{32}{\sqrt{29}} \end{aligned}$$

Example 73. *Find the directional derivative* $D_{\mathbf{u}} f(x,y)$ *if* $f(x,y) = x^3 - 3xy + 4y^2$ *in the direction of angle* $\theta = \frac{\pi}{6}$ *to the x-axis. Also find* $D_{\mathbf{u}} f(1,2)$.

We have that $\mathbf{v} = (\sqrt{3}, 1)$ because trigonometry (bro). Thus

$$\begin{aligned} D_{\mathbf{u}} f(x,y) &= \nabla f(x,y) \cdot \mathbf{u} \\ &= \langle 3x^2 - 3y, -3x + 8y \rangle \cdot \frac{\langle \sqrt{3}, 1 \rangle}{\| \langle \sqrt{3}, 1 \rangle \|} \\ &= \langle 3x^2 - 3y, -3x + 8y \rangle \cdot \left\langle \frac{\sqrt{3}}{2}, \frac{1}{2} \right\rangle \\ &= \frac{1}{2}(3\sqrt{3}x^2 - 3x + (8 - 3\sqrt{3})y) \end{aligned}$$

Plugging in our values for $x = 1$ and $y = 2$, we have

$$\begin{aligned} D_{\mathbf{u}} f(1,2) &= \frac{1}{2}(3\sqrt{3}(1)^2 - 3(1) + (8 - 3\sqrt{3})(2)) \\ &= \frac{1}{2}(13 - 3\sqrt{3}). \end{aligned}$$

Theorem 13. The maximum value of $D_{\mathbf{u}} f(x,y)$ is $\|\nabla f(x,y)\|$.

Proof.

$$\begin{aligned} D_{\mathbf{u}} f(x,y) &= \nabla f \cdot \mathbf{u} \\ &= \|\nabla f(x,y)\| \|u\| \cos\theta \\ &= \|\nabla f(x,y)\| \cos\theta. \end{aligned}$$

This value is maximized when $\theta = 0$, or when $\cos\theta = 1$. □

Example 74. *Find the maximum value of* $D_{\mathbf{u}} f(x,y)$ *of the previous problem.*

$$\begin{aligned} \nabla f(x,y) &= \langle 3x^2 - 3y, -3x + 8y \rangle \\ \nabla f(1,2) &= \langle -3, 13 \rangle. \end{aligned}$$

So the maximum value is

$$\|\nabla f(1,2)\| = \sqrt{(-3)^2 + (13)^2} = \sqrt{178}.$$

The unit vector $\mathbf{u}$ that gives the maximum is $\frac{\langle -3,13\rangle}{\sqrt{178}}$.

Additional Examples

Example 75. Let $w = e^{xy+xz+yz}$ where $x = 3u+v, y = 3u-v$, and $z = u^2v$. Find

$$\left.\frac{\partial w}{\partial v}\right|_{(u,v)=(0,1)}.$$

$$\begin{aligned}\frac{\partial w}{\partial v} &= \frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial v} + \frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial v} + \frac{\partial w}{\partial z}\cdot\frac{\partial z}{\partial v}\\ &= (e^{xy+xz+yz}\cdot(y+z))\cdot(1) + (e^{xy+xz+yz}\cdot(x+z))\cdot(-1) + (e^{xy+xz+yz}\cdot(x+y))\cdot(u^2)\\ &= e^{xy+xz+yz}\cdot(y-x+xu^2+yu^2)\end{aligned}$$

When $u=0, v=1$, we have

$$x = 1, \quad y = -1, \quad z = 0.$$

Plugging this in, we have

$$e^{(1)(-1)+(1)(0)+(-1)(0)}\cdot((-1)-(1)) = -2e^{-1}$$

Example 76. *Let $f(x,y,z) = z^4 - x^3y^2$. Find the directional derivative of f at the point $P(1,3,2)$ in the direction of the vector from P to $Q(2,4,3)$.*

First, we find the vector $\overrightarrow{PQ}$

$$\overrightarrow{PQ} = Q - P = \langle 1,1,1\rangle.$$

Then, we have

$$\mathbf{u} = \frac{\langle 1,1,1\rangle}{\sqrt{3}}.$$

To calculate the directional derivative

$$\begin{aligned}D_{\mathbf{u}}f(x,y,z) &= \nabla f(x,y,z)\cdot\mathbf{u}\\ &= \langle -3x^2y^2, -2x^3y, 4z^3\rangle\cdot\left\langle\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right\rangle\end{aligned}$$

Plugging in $(x,y,z) = (1,3,2)$, we have

$$\begin{aligned}\langle -3(1)^2(3)^2, -2(1)^3(3), 4(2)^3\rangle\cdot\left\langle\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right\rangle &= \frac{-27}{\sqrt{3}} + \frac{-6}{\sqrt{3}} + \frac{24}{\sqrt{3}}\\ &= \frac{-1}{\sqrt{3}}.\end{aligned}$$

Tangent Planes

Given a surface S with equation $z = f(x,y)$, we find the *tangent plane* to S at a given point P on the surface. To be precise, we consider any two curves C_1, C_2 on the surface S that pass through the point P. Let T_1 and T_2 be the tangent line of these curves at the point P. The tangent plane is the plane that contains the lines T_1 and T_2.

Assume the tangent plane passing through the point $P(x_0, y_0, z_0)$ to be

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0,$$

where $\langle A, B, C\rangle$ is the normal vector $\mathbf{n}$ of this plane. We then do implicit differentiation of this equation with respect to x and y to get

$$A = -Cf_x, \quad B = -Cf_y.$$

Plug these back and simplify (assume $C \neq 0$) to get the formula of the tangent plane:

$$z = z_0 + f_x(x - x_0) + f_y(y - y_0).$$

Example 77. *Find the tangent plane to the elliptic paraboloid $z = 2x^2 + y^2$ at the point* $(1,1,3)$.

We have that

$$f_x = 4x, \quad f_y = 2y.$$

Plugging in at the point when $(x,y) = (1,1)$, we have that

$$f_x(1,1) = 4, \quad f_y(1,1) = 2.$$

And thus, the tangent plane is

$$z = 3 + 4(x-1) + 2(y-1) \implies 4x + 2y - 3 = z.$$

Tangent Planes of Implicit Functions

If we are given a surface $F(x,y,z) = c$, we want to find the tangent plane at the point $P(x_0, y_0, z_0)$. Consider the curve C on this surface, $\mathbf{r}(t) = \langle x(t), y(t), z(t)\rangle$. After differentiating on both sides with respect to t, we have that

$$\nabla F \cdot \mathbf{r}'(t) = 0.$$

This means that ∇F is perpendicular to any tangent vector at point P, and thus ∇F is the normal vector of the tangent plane. This means that

$$F_x(x_0,y_0,z_0)(x - x_0) + F_y(x_0,y_0,z_0)(y - y_0) + F_z(x_0,y_0,z_0)(z - z_0) = 0.$$

Example 78. *Given the ellipsoid $x^2 + 4y^2 + z^2 = 18$*

(a) Find the tangent plane to this ellipsoid at the point $(1,2,1)$.

(b) Find the parametric equations of the normal line (line perpendicular to the plane) at this point.

(c) Find the acute angle that the plane makes with the xy-plane.

(a) First we find the partial derivatives

$$F_x = 2x, \quad F_y = 8y, \quad F_z = 2z.$$

Next we plug in our points

$$F_x(1,2,1) = 2, \quad F_y(1,2,1) = 16, \quad F_z(1,2,1) = 2.$$

The tangent plane is thus

$$\begin{aligned} F_x(1,2,1)(x-1) + F_y(1,2,1)(y-2) + F_z(1,2,1)(z-1) &= 0 \\ 2(x-1) + 16(y-2) + 2 &= 0 \\ 2x - 2 + 16y - 32 + 2z - 2 &= 0 \\ 2x + 16y + 2z &= 36 \\ x + 8y + z &= 18. \end{aligned}$$

(b) We have the normal vector $\mathbf{n} = \langle 1,8,1 \rangle$ so the parametric equations of the normal line at $(1,2,1)$ are

$$x = 1 + t, \quad y = 2 + 8t, \quad z = 1 + t.$$

(c) The angle between the two planes is the same as the angle between their normal vectors. First, let's find the normal vector of the xy-plane, which is just the z-axis.

$$\begin{aligned} \cos\theta &= \frac{\langle 1,8,1 \rangle \cdot \langle 0,0,1 \rangle}{\|\langle 1,8,1 \rangle\| \|\langle 0,0,1 \rangle\|} \\ &= \frac{1}{\sqrt{66}} \\ \theta &= \cos^{-1}\left(\frac{1}{\sqrt{66}}\right). \end{aligned}$$

Remark 1. For the special case when the surface is written in the form $z = f(x,y)$, we can rewrite the relation as

$$0 = F(x,y,z) = z - f(x,y).$$

We apply the previous formula to get

$$-f_x(x_0,y_0)(x-x_0) - f_y(x_0,y_0)(y-y_0) + (z-z_0) = 0,$$

or

$$z = z_0 + f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0),$$

which agrees with the first formula of this section.

Maxima and Minima of Functions of Two Variables

Theorem 14. If f has a local maximum or minimum at (a,b) and the first-order partial derivatives of f exist there, then $f_x(a,b) = 0$ and $f_y(a,b) = 0$. A point (a,b) is called a critical point.

Example 79. *Find all the critical points of* $f(x,y) = x^2 + y^2 - 2x - 6y + 14$.

We take the partial derivatives and solve

$$f_x = 2x - 2 = 0 \implies x = 1,$$

and also

$$f_y = 2y - 6 = 0 \implies y = 3.$$

Thus, $(x,y) = (1,3)$ is a critical point of f.

Second Derivative Test

Theorem 15 (Second Derivative Test). Suppose that the second partial derivatives of f are continuous on a disk with center (a,b), and suppose that $f_x(a,b) = 0$ and $f_y(a,b) = 0$. Let D be defined as

$$D = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix}.$$

Then

1. If $D > 0$ and $f_{xx}(a,b) > 0$, then $f(a,b)$ is a local minimum.
2. If $D > 0$ and $f_{xx}(a,b) < 0$, then $f(a,b)$ is a local maximum.
3. If $D < 0$ then $f(a,b)$ is not a local maximum or minimum.

We call the point (a,b) in case 3 a saddle point.

Example 80. *Find the local maximum, local minimum and saddle points of*

$$f(x,y) = x^4 + y^4 - 4xy + 1.$$

First, we find the critical points

$$\begin{aligned} f_x &= 4x^3 - 4y \\ f_y &= 4y^3 - 4x. \end{aligned}$$

To determine the critical values, we solve the system of equations

$$x^3 - y = 0 \tag{1}$$
$$y^3 - x = 0 \tag{2}$$

This implies that $y = x^3$ and $x = y^3$. Plugging y into (2), we have

$$\begin{aligned} 0 &= 4(x^9 - x) \\ &= 4x(x^8 - 1) \\ &= 4x(x^4 - 1)(x^4 + 1) \\ &= 4x(x^4 + 1)(x^2 + 1)(x + 1)(x - 1). \end{aligned}$$

Thus, $x = 0, 1, -1$ and from this $y = 0, 1, -1$. So the critical points are

$$(x, y) = (0,0), (1,1), (-1,-1).$$

After obtaining the critical values, we apply the second derivative test. First, let's compute the second derivatives

$$\begin{aligned} f_{xx} &= 12x^2 \\ f_{xy} &= -4 \\ f_{yx} &= -4 \\ f_{yy} &= 12y^2. \end{aligned}$$

So then, at $(0,0)$

$$\begin{aligned} D &= \begin{vmatrix} 12(0)^2 & -4 \\ -4 & 12(0)^2 \end{vmatrix} \\ &= \begin{vmatrix} 0 & -4 \\ -4 & 0 \end{vmatrix} \\ &= -16 < 0. \end{aligned}$$

Thus, $(0,0)$ is a saddle point. At $(1,1)$

$$\begin{aligned} D &= \begin{vmatrix} 12 & -4 \\ -4 & 12 \end{vmatrix} \\ &= 144 - 16 \\ &= 128 > 0. \end{aligned}$$

Since $f_{xx}(1,1) = 12 > 0$, we have that $f(1,1)$ is a local minimum. At $(-1,-1)$

$$\begin{aligned} D &= \begin{vmatrix} 12 & -4 \\ -4 & 12 \end{vmatrix} \\ &= 128 > 0. \end{aligned}$$

Since $f_{xx}(-1,-1)$ is also positive, we have that $f(-1,-1)$ is also a local minimum.

Example 81. *Find the shortest distance from the point* $(1,0,-2)$ *to the plane* $x + 2y + z = 4$.

This is a minimization problem with the constraint that a point has to be on the plane

$$\begin{aligned} D &= \sqrt{(x-1)^2 + (y-0)^2 + (z+2)^2} \\ D &= \sqrt{(x-1)^2 + (y-0)^2 + (6-x-2y)^2} \\ D^2 &= (x-1)^2 + (y-0)^2 + (6-x-2y)^2 \end{aligned}$$

We now have $f(x,y) = D^2$. We want to solve

$$\begin{aligned}
0 &= f_x \\
&= 2(x-1) + 2(6-x-2y)(-1) \\
&= 2(2x+2y-7) \\
0 &= f_y \\
&= 2y + 2(y-x-2y)(-2) \\
&= 2(2x+5y-12).
\end{aligned}$$

So we have a system of equations for two variables

$$\begin{aligned}
2(2x+2y-7) &= 0 \\
2(2x+5y-12) &= 0
\end{aligned}$$

Solving this, we have

$$(x,y) = \left(\frac{11}{6}, \frac{5}{3}\right).$$

Thus the shortest distance is

$$\begin{aligned}
D &= \sqrt{\left(\frac{11}{6}-1\right)^2 + \left(\frac{5}{3}-0\right)^2 + \left(6-\frac{11}{6}-2\left(\frac{5}{3}\right)\right)^2} \\
&= \frac{5\sqrt{6}}{6}.
\end{aligned}$$

We can also verify using the formula where $(x_0, y_0, z_0) = (1, 0, -2)$

$$\begin{aligned}
D &= \frac{|ax_0 + by_0 + cz_0 - d|}{\sqrt{a^2+b^2+c^2}} \\
&= \frac{5}{\sqrt{6}} \\
&= \frac{5\sqrt{6}}{6}
\end{aligned}$$

Absolute Maximum

The absolute maximum/minimum are the points that can be found from a set of all the critical points and points on the boundary. The method for finding them is as follows

1. Find the values of f at the critical points of f inside the region.
2. Find the extreme values of f on the boundary.
3. The largest value from step 1 and step 2 is the absolute max value. The smallest value from step 1 and step 2 is the absolute minimum value.

Example 82. *Find the absolute maximum and minimum values of* $f = 2x^2 + y^3$ *where the region* $D = \{(x,y) \mid x^2 + y^2 \leq 1\}$.

We have that D is a disk of radius 1.

1. First, we find the values of the critical points inside the region.

$$f_x = 4x = 0 \implies x = 0$$
$$f_y = 3y^2 = 0 \implies y = 0.$$

We have that $(x, y) = (0, 0)$ is a critical point.

2. The boundary of D is $x^2 + y^2 = 1$, so plugging into the function, we have

$$\begin{aligned} f &= 2x^2 + y^3 \\ &= 2(1 - y^2) + y^3 \\ &= 2 - 2y^2 + y^3. \end{aligned}$$

Next we calculate the partial derivatives for the boundary (we consider only w.r.t y since there's only y)

$$\begin{aligned} \frac{df}{dy} &= -4y + 3y^2 \\ &= y(-4 + 3y). \end{aligned}$$

So $y = -\frac{4}{3}, 0$. When $y = 0$ we have that $x = \pm 1$. When $y = -\frac{4}{3}$, there is no solution for x. Thus, we have the critical points

$$(x, y) = (1, 0), (-1, 0).$$

We can conclude that $(0,0), (1,0), (-1,0)$ are the three candidates for absolute max and absolute min.

3. We can now compare each candidate by plugging them into the original function

$$\begin{aligned} f(0,0) &= 0 \\ f(1,0) &= 2 \\ f(-1,0) &= 2. \end{aligned}$$

We see that the $f(0,0)$ is the absolute minimum while $f(1,0)$ and $f(-1,0)$ are the absolute maxima.